Personal and Political Agency Across the Urban-Rural Gradient

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This study uses World Values Survey 7 (2017-2023) across 48-61 countries, depending on specification, to test urban-rural differences in personal agency and political agency. Our findings indicate that personal agency is generally not higher in urban than in rural areas, contrary to the expectation of the "city air makes you free" hypothesis–in most specifications, we find little to no difference. However, political agency is significantly higher in urban than in rural areas in high-income countries, consistent with the "left behind ruralities" hypothesis. Furthermore, as Toennies hypothesized, those doing well financially are "active and alive" in cities or, as we find here, have greater personal agency (but not political agency). Effect sizes are small for personal agency, but considerable for political agency–in high income countries political agency adjusted for its predictors increases from <20k to >500k by about .2 on 1-5 scale. As in any correlational study, these findings do not establish causality.

World Values Survey, freedom, agency, autonomy, personal agency, political agency, urbanism, urban, rural, urban-rural, cities

"City life and Gesellschaft down the common people to decay and death." Ferdinand Toennies

1 Introduction

"City air makes you free"[1]expresses a longstanding belief that cities offer freedom from the social constraints of rural life (Yamagishi et al. 2012). The saying evokes the expectation that cities provide greater opportunities for individuals to shape their own lives. In the social sciences, and sociology in particular, urbanization has long been viewed as one of the major drivers of human progress and empowerment. Classical sociologists described how the concentration of people in cities facilitates and necessitates a finer division of labor and greater specialization to satisfy existing and newly emerging increasingly diverse needs (Durkheim [1988] 2016, Simmel [1901-1908]1995). Urbanization is accompanied by and interwoven with other large-scale processes such as economic development, industrialization, and social mobility, which increase the number of social interactions and interdependencies while diversifying society and weakening collective control over the individual (Durkheim [1988] 2016, Simmel [1901-1908]1995). Cities are a breeding ground for individual freedom–there individuals more often interact with and depend more strongly on people with different occupations, lifestyles, and values, while also enjoying greater freedom themselves to pursue different life paths in domains such as work, relationships, family, and leisure.

At the same time, the classical sociologists already warned against the relentless expansion of needs, whereby the city evokes wants and needs stimulated by ever-new products, services, and experiences, while people simultaneously become increasingly indifferent to these stimuli and to the novelty they offer (Durkheim [1988] 2016, Simmel [1901-1908]1995). Urban life may thus expand individuals' personal agency, i.e., their freedom to shape their lives in accordance with their own goals and values, but it may also constrain it by drawing them ever deeper into an endless cycle of expanding needs (Schwartz 2004). Cities may offer more choices in employment, relationships, and lifestyles, but whether these possibilities translate into greater personal agency remains an empirical question. To

[1] "Stadt Luft macht frei" (Park et al. [1925] 1984, p. 12). This saying originated in the Middle Ages, and it meant freedom from feudalism in cities; cities were non-feudal islands in a sea of feudalism (Harvey 2012). Originally, the saying referred to freedom from feudal obligations (Harvey 2012, Park et al. [1925] 1984, Yamagishi et al. 2012).

date, surprisingly little is known about the urban-rural divide in personal agency, and existing research has yielded mixed results (e.g., Atherton et al. 2024a, Kusnali et al. 2025a, Lolle 2023).

Urbanization may also shape political agency, that is, individuals' confidence in their ability to influence political decisions and thereby shape their country and its future. In their classic account of the origins of modern party systems, Lipset and Rokkan (1967) identify the urban-rural cleavage as one of four fundamental social cleavages, rooted in the conflict between agrarian interests in the countryside and emerging industrial and commercial interests in the cities. Historically, this conflict found expression in the formation of parties representing rural and agrarian interests, on the one hand, and urban and industrial interests, on the other. Although the political significance of this cleavage has changed over time, the urban-rural divide remains visible in contemporary electoral politics, for instance in differences in support for Green and far-right parties (e.g., Crulli 2024, Marks et al. 2020, Rickardsson 2021).

"Left behind ruralities" hypothesis refers to rural resentment, feelings of powerlessness, and political inefficacy. Few studies directly quantify the urban-rural gap in perceived political agency. The few that do, however, find that urban residents perceive greater political agency than their rural counterparts (e.g., García Del Horno et al. 2024, Luukkonen et al. 2022, Rowland et al. 2024).

These differences may partly reflect the unequal distribution of economic resources and opportunities across urban and rural areas. Both personal and political agency are substantially shaped by prevailing economic conditions and individual material resources. At the societal level, economic progress can expand people's opportunities by improving infrastructure, services, and welfare provision. At the individual level, it can reduce the most existential constraints on human choice by increasing access to, and the availability of, resources that provide the objective means to exercise choice (Welzel and Inglehart 2001, Welzel et al. 2003).

Although economic progress is ultimately expected to expand human choice and opportunities for all, urban-rural disparities in these resources and opportunities tend to be substantially wider in developing than in developed countries, as has been documented, for instance for employment and wages (Ananian and Dellaferrera 2024), health care (Marthias et al. 2025), and life satisfaction (Easterlin et al. 2011). Further, urban-rural differences in outcomes are partly attributable to inequalities in individual resources such as income, as studies of right-wing voting (Rickardsson 2021) and life satisfaction (Easterlin et al. 2011) demonstrate.

The greater range of choices available in cities does not guarantee that residents possess the means to exercise them. Thus, urbanization may expand the possibilities for personal agency without generating the same perceived freedom for everyone.

This suggests that the opportunities associated with urban life are not equally accessible to all, echoing Tönnies's observation that "[...] in the city and therefore where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive" (Tönnies [1887] 2002, p. 227).

These arguments raise three related questions: Do urban residents perceive greater personal agency and political agency than rural residents? Do these differences vary with countries' levels of economic development? And does individual material well-being shape whether urban opportunities translate into greater perceived agency?

To date, only a few studies have investigated urban-rural divides in personal and political agency, and they have predominantly focused on Western contexts and often focus on only one country. Those covering multiple countries have generally not examined cross-national variation or differences between social groups. Our paper thus makes three main contributions to the literature. First, we examine urban-rural divides in personal and political agency across a more heterogeneous set of countries than previous studies and explore how these divides vary between countries. Second, we investigate whether urban-rural divides in personal agency and political agency vary systematically with countries' levels of economic development. Third, we examine whether these divides vary according to individuals' satisfaction with material resources.

In what follows, we first review the theoretical expectations concerning urban-rural differences in personal agency and political agency. We then describe the data and analytical approach, present our results, and consider their implications and limitations.

2 Literature

2.1 Perceived Personal Agency: "City Air Makes You Free" Hypothesis

What is "perceived personal agency" or "freedom" for short?[2] To be free is to be able to choose, and to control and direct one's own life. Freedom is having control–a person is free if she is her own master. It could be labeled 'psychological freedom': "Freedom means a degree of harmony between basic motives and overt behavior" (Bay 1970, p. 83 cited in Okulicz-Kozaryn 2014). It is the "capability to choose," having "awareness of alternatives" and "courage to choose" (Veenhoven 2000).[3] Beyond its potential benefits for individual well-being, freedom also has intrinsic value. The Universal Declaration of Human Rights recognizes freedom as a human right essential for the realization of one's human potential (arts. 26, 29). Personal freedom has been related to a multitude of positive outcomes such as better physical and mental health and better subjective well-being. Yet freedom also has value in its own right. Cities offer more choices than rural areas and, in this sense, more freedom. Yet, it is important to underscore that too much choice may also backfire and make it difficult to act or to choose, possibly leading to paralysis (Schwartz 2004). Freedom is a broad concept encompassing multiple dimensions of individuals' ability to make choices and shape their lives. In this study, we focus specifically on perceived personal agency. While closely related to freedom, personal agency captures one particular dimension rather than the concept in its entirety. Hitlin and Long (2009) review the concept in sociology (see also D'Italia and Okulicz-Kozaryn 2025).

There are theoretical reasons to expect more personal agency in cities than in rural areas. In rural areas, mechanical solidarity prevails–rural residents function like parts of a simple machine–groups are held together through homogeneity and similarity: similar lifestyles, roles, shared values, and morality (Durkheim [1893] 1997). Hence, rural areas enforce social conformity that is not conducive to individual freedom. Cities provide freedom from traditional rural social constraints–Gesellschaft versus Gemeinschaft–deviation from community norms is more likely to be reprimanded and ostracized in rural than urban areas. On the other hand, organic solidarity in cities allows for more differentiation and individualism. An urbanite may be more internally driven than rural residents, who are more externally constrained (Yamagishi et al. 2012). While urbanites are more tolerant of others (Tuch 1987), they are also more reserved. Simmel (1903) observed that urbanism forces individual differentiation: in an environment characterized by diversity and constant stimulation, individuals must distinguish themselves from others through increasingly extreme means, including extravagance and exaggeration. To some extent, this pressure to stand out may foster differentiation, individualism, and personal freedom. At the same time, the overload of stimuli overwhelms individuals, prompting them to distance themselves from their environment and adopt a blasé attitude toward others (Simmel 1903). Such responses may protect individuals' psyches, but emotional detachment and social withdrawal are not expressions of agency.

Cities provide more employment and social opportunities, so an urbanite is freer by being less concerned about social exclusion as she has plenty of alternatives (Yamagishi et al. 2012).[4] One mechanism is higher social pressure to conform in rural areas–cities provide anonymity–indeed, concern for moral reputation is greater in rural than urban areas (Okulicz-Kozaryn 2026). The higher the population density, the lower the frequency of social comparison (Okulicz-Kozaryn 2026).

But theoretical arguments also suggest that people may feel less free in cities. City life is stressful, hectic, anxiety-provoking, materialistic, consumerist (Okulicz-Kozaryn 2015), and overstimulating (Simmel 1903, Lederbogen et al. 2011)–under such conditions, feeling free can be difficult. Steve Pile in his colorful writing argues that the history and past of old cities can be a weight on their inhabitants (Pile 2005a,b, Pile et al. 1999), and old cities carry melancholia (Pile 2005b). These perspectives suggest that urban areas can inhibit how one perceives personal agency, even as cities expand opportunities for individual choice.

A multitude of studies have examined individual predictors of personal agency (e.g., Okulicz-Kozaryn 2014, Pitlik 2016, Nikolaev and Bennett 2016), but none of them examine urbanicity. The differences in freedom across the urban-rural gradient have been

[2] Here we focus on personal as opposed to agency/societal freedom and follow Okulicz-Kozaryn (2014) in this paragraph.

[3] The degree of perceived control over choice is called "locus of control" in psychology and Verme (2009) argues that the WVS item used here measures locus of control in addition to freedom.

[4] Also see discussion in Luca et al. (2023).

examined in only a handful of studies covering very few countries. These studies offer a mixed picture, examining both general perceptions of agency and control over specific aspects of daily life. Lolle (2023) finds no differences in agency across communities of different sizes in Denmark. Atherton et al. (2024b) find that psychological well-being–which entails agency as a sub-component–is lower in the most rural places compared to the most urban places in the USA; however, these effects become statistically insignificant once socio-demographic factors are controlled for. Kusnali et al. (2025b) find that women in Indonesia have more decision-making agency in many (i.e., household expenditures, food expenditures, time allocation to social activities) but not all (e.g., contraceptive use) aspects of daily life in urban compared to rural areas. Indian women have more autonomy in cities than in rural areas, both in personal and household decisions (Mishra 2014). Båge et al. find no urban-rural difference in people's perceived reproductive agency, i.e., how much freedom of choice and control they have over family planning, in Ethiopia, Kenya, and Zimbabwe (Båge et al. 2024) and sub-Saharan Africa (Båge et al. 2026). Taken together, these findings leave open the question of whether urban-rural differences in perceived personal agency follow a universal pattern across countries.

2.2 Perceived Political Agency: "Left Behind Ruralities" Hypothesis

As the world continues to urbanize, rural areas are disappearing (Pospěch et al. 2024). Rurality in the developed world is increasingly peripheral, with declining political power and voice relative to urban centers and being abandoned and disrespected–thereby fueling rural discontent, resentment, and shame (see accounts in Hanson 2015, Fuller 2017). It is therefore hardly surprising that the more urbanized affluent democracies are increasingly witnessing a growing political divide between urban and rural areas, evident in voting patterns (Huijsmans and Rodden 2025, Mettler and Brown 2022), political trust (Mitsch et al. 2021), and perceptions of individual and community recognition (Hyland et al. 2024).

Rural resentment can emerge when people feel that policy is imposed on them, rather than developed with or for them; and there is racial resentment where rural Whites resent urban minorities and immigrants getting better treatment than them (Diamond 2023). Although rural life involves reliance on community and tight connections, self-reliance, autonomy, and freedom also remain core values (Diamond 2023).

Rural areas are 'left-behind' (Ulrich-Schad and Duncan 2021) and 'places that don't matter' (Rodríguez-Pose 2018)—they are second-tier areas with limited or deteriorating opportunities. When people's opportunities are constrained, their freedom to exercise agency is likewise restricted (Ryan and Deci 2004, Welzel 2013). Exercising agency requires access to meaningful and substantively different opportunities from which individuals can choose in pursuing lives that are consistent with their own goals and values (Sen 1993, 1999, 2005). Limited opportunities may therefore undermine both people's perceived ability to shape their own lives in accordance with these goals and values, and their perceived ability to shape the conditions in which their lives are embedded, such as institutions, infrastructure, and the built and natural environment. We therefore expect people living in 'left-behind' rural areas to perceive themselves as having less influence over the conditions in which they are embedded and that shape their lives, and consequently to perceive lower levels of political agency.

Only a few studies have investigated urban-rural differences in political agency. Studies investigating the urban-rural divide in political agency consistently find that people in urban areas perceive themselves as more politically efficacious than those in rural areas, both in Europe (García Del Horno et al. 2024, Luukkonen et al. 2022) and the United States (Rowland et al. 2024).

Exploring rural resentment in Europe and the United States, Margalit et al. (2025) find that rural residents who feel they have little or no say in what the government does are more likely to vote for populist parties in Europe and for Trump in the United States. Political agency has been shown to be positively linked to political engagement (Accetta 2023).[5]

Overall, these studies suggest that rural residents may feel less political agency than urban residents. However, to our knowledge, no study has investigated whether this relationship is universal across different contexts (such as countries with high or low GDP).

[5]Self-efficacy and political agency have been shown to have opposing effects on the justification of terrorism and political violence: people who perceive themselves as highly self-efficacious and those who feel they have little political agency are less likely to justify such actions (Julkif 2022).

3 Hypotheses

Against this backdrop, we investigate the following hypothesis:

H1: Respondents residing in urban areas report significantly higher levels of personal and political agency than those residing in rural areas.

Given that economic and social progress unfolds unevenly across urban and rural areas depending on a country's level of development, we further hypothesize that the urban-rural divide is more pronounced in less developed countries:

H2: The urban-rural divide in personal and political agency is significantly larger in low- and middle-income countries than in high-income countries.

Finally, as differences in individual material resources may account for part of the urban-rural divide, we examine whether its magnitude varies across levels of individual material resources:

H3: The urban-rural divide in personal and political agency is smaller at higher levels of individual material resources.

A variation on this hypothesis, following Toennies' quote, is that the greater agency observed in cities mostly stems from well-to-do urbanites.

4 Data and analytical strategy

To test these hypotheses, we use data from Wave 7 of the World Values Survey (Haerpfer 2022). The WVS is a repeated cross-sectional survey conducted approximately every five years since 1981 to study social, political, economic, religious, and cultural values, beliefs, and norms worldwide (World Values Survey 2026a,b). It draws representative samples of each country's adult population using simple or multistage random sampling, depending on the country (World Values Survey 2026c). Data for Wave 7 were collected between 2017 and 2023, primarily through face-to-face interviews, although other modes of data collection, such as telephone interviews and self-completion, were also used in some countries (World Values Survey 2026c).

4.1 Operationalization

Urban-Rural. Although strictly speaking, urban-rural is a gradient rather than a dichotomy (Okulicz-Kozaryn 2017), we employ a categorical operationalization of the degree of urbanization. This approach is consistent with the theoretical assumptions discussed previously, which commonly postulate a rift between the rural countryside on the one hand and the metropolis on the other. Additionally, as survey respondents are often unevenly distributed across the community-size categories captured in surveys, collapsing these categories is common practice in empirical comparative urban-rural research (Okulicz-Kozaryn and Valente 2021). The original town-size variable, coded by interviewers into eight categories ranging from fewer than 2,000 to 500,000 or more inhabitants, was thus collapsed into three categories: (1) rural areas, comprising communities with fewer than 20,000 inhabitants; (2) towns, comprising communities with 20,000 to fewer than 500,000 inhabitants; and (3) cities, comprising communities with 500,000 or more inhabitants.

In the appendix we do several robustness checks, including dropping countries with fewer than 50 observations per category extreme category. Results are similar except for interactions with financial satisfaction where we drop if town size in the <20k people category had <50obs (results are also robust when dropping countries where the >500k residents category had <50obs).

Personal agency. We operationalize personal agency using respondents' self-assessed freedom of choice and control, measured with the following item: "Some people feel they have completely free choice and control over their lives, while other people feel that

what they do has no real effect on what happens to them. Please use this scale where 1 means 'no choice at all' and 10 means 'a great deal of choice' to indicate how much freedom of choice and control you feel you have over the way your life turns out." Higher values indicate stronger perceived personal agency. This item is commonly used to capture individuals' perceived agency (e.g., Verme 2009, Inglehart et al. 2008, Veenhoven 2000, Okulicz-Kozaryn 2014) and related concepts such as autonomy, locus of control, and fatalism (see Kouba 2016) for a conceptual discussion and (Steckermeier 2022) for a review of empirical research based on this item).

All variables including controls are listed in the appendix and the distribution of all variables are also reported. We control for married, employment type dummies (full/part time, self-employed, retired, housewife, student, unemployed, and other), financial satisfaction,[6] health, kids, and importance of religion.

Additional robustness tests were conducted resulting in oversaturated models; the addition of these controls yields larger coefficients, although overall substantive conclusions remain the same. Refer to the Appendix for more detail.

Political agency. We operationalize political agency via respondents' answers to the question: "How much would you say the political system in your country allows people like you to have a say in what the government does?" Responses range from 1 ("A great deal") to 5 ("Not at all"). We reverse the response categories so that higher values indicate stronger perceived political agency. The item is regularly used to capture external political agency (e.g., Sumino 2025, Bienstman et al. 2024) and related concepts such as having a political voice (Prats et al. 2023) and perceptions of political unfairness (Im 2024). Unfortunately, this item was not fielded in Bolivia, Chile, China, Germany, Ecuador, Greece, South Korea, Puerto Rico, Romania, Russia, Serbia, Thailand, Turkey, and the United States. The sample for this part of the analysis is thus reduced to 48 countries (compared to 61 countries in the analysis of personal agency).

Control variables. In all analyses pertaining to political agency we control for the following factors which have been shown to be associated with political agency as well as to differ between urban and rural areas: `education` (political agency: (Emmenegger et al. 2015, García Del Horno et al. 2024, Marx and Nguyen 2016, Rowland et al. 2024) urban-rural: (Van Maarseveen 2021, Zahl-Thanem and Rye 2024), `employment` (political agency: Emmenegger et al. (2015), Luukkonen et al. (2022), Marx and Nguyen (2016) urban-rural: Eurostat (2026), Ananian and Dellaferrera (2024), `material resources` (political agency: Emmenegger et al. (2015), Luukkonen et al. (2022), Marx and Nguyen (2016), Rowland et al. (2024); urban-rural: Finnemann et al. (2024), Kpatcha et al. (2026), Meloni et al. (2024), `health` (political agency: García Del Horno et al. (2024), Gidengil and Wass (2024), Shore et al. (2019); urban-rural: MacKinnon et al. (2023), Menashe-Oren and Masquelier (2022), and `religiosity` (political agency: Marx and Nguyen (2016), Rowland et al. (2024) ; urban-rural: Balazka (2020), Meulemann and Schmidt-Catran (2020).

In all analyses pertaining to personal agency we additionally control for having a family because living with a partner and/or children has been shown to be associated with personal agency (e.g., Nikolaev and Bennett 2016, Steckermeier 2024, Symoens et al. 2014) as well as to differ between urban and rural areas (e.g., Gautier et al. 2010, Stockdale and Catney 2014). It is worth emphasizing that a lot of studies that investigate antecedents of personal agency–and none of them look at urban-rural differences.

5 Results

5.1 Interactions with Level of Development

We begin by reporting in table 1 mean personal agency and political agency by settlement size and level of development.

[6] Financial satisfaction is key–not everyone needs the same amount of money to be financially satisfied and feel free. This is measured as "How satisfied are you with the financial situation of your household?"

Table 1: Means by level of development. Categorization of development according to World Bank income groups. LIC/LMIC=Lower income countries and lower middle income countries. UMIC=Upper middle income countries. HIC=High income countries.

	personal agency				**political agency**			
		country development				country development		
place	overall	LIC/LMIC	UMIC	HIC	overall	LIC/LMIC	UMIC	HIC
<20k	7.20	7.11	7.26	7.24	2.96	3.03	3.06	2.67
20-500k	7.27	7.21	7.21	7.36	2.88	2.95	2.83	2.87
>500k	7.12	6.96	7.40	6.98	2.89	2.67	2.86	2.99

Table 1 shows the average reported levels of personal and political agency across rural, town, and city areas, by societal level of development and individual-level material resources.

Overall, personal agency is lowest in cities and highest in towns, with rural areas in between. This pattern is also observed in less developed countries and especially high-income countries, but not in upper-middle-income countries, where the ranking is reversed– levels are highest in cities. These differences, however, are small and disappear when controlling for the above-mentioned control variables (see "Variables' Definitions and Descriptive Statistics" in the Supplementary Online Material (SOM)).

Political agency, by contrast, is highest in rural areas, followed by cities and towns, which show practically identical levels. However, patterns of political agency differ considerably by a country's level of development. In less developed countries, average perceived political agency is highest in rural areas and lowest in cities. In high-income countries, this pattern is reversed: people in cities report higher political agency than those in towns, who in turn report higher agency than those in rural areas. In upper middle, the pattern largely mirrors the overall results, with political agency highest in rural areas, followed by cities and then towns, where levels are relatively similar.

This difference holds even when controlling for predictors of political agency (see SOM). Margins are plotted in figure 1.

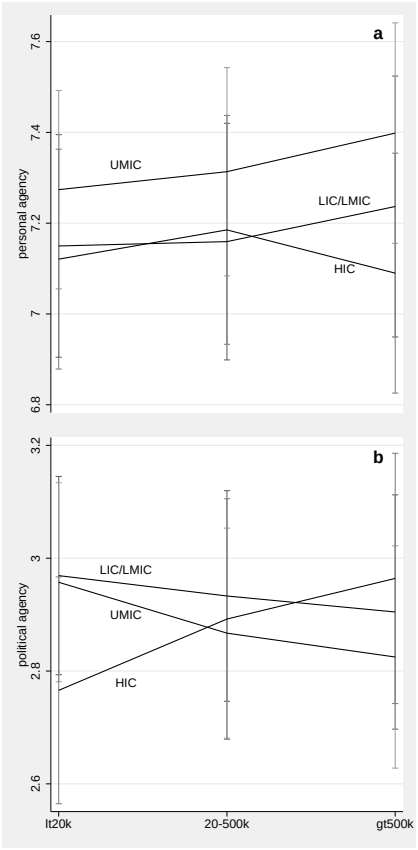


Figure 1: Predicted `personal agency` in panel a, and `political agency` in panel b on y-axes across rural-urban on x-axis by level of development. `Personal agency` adjusted for married, employment status, financial satisfaction, health, children, and religion important. `Political agency` adjusted for education, employment status, financial satisfaction, health, and religion important. Regressions are in appendix. These results are robust when removing countries with less than 50 observations in rural or city areas (see Appendix)."

In a nutshell, no estimates are significant for personal agency except one marginally significant, but several models are significant for political agency. Margins with `pwcompare` are in the appendix. `Personal Agency:` With one exception, there are no significant differences in the level of perceived agency between rural, town, and city areas irrespective of the level of development: In In Upper Middle Income Countries (UMIC), people in city areas report significantly more personal agency than in rural areas (b=0.12*). `Political Agency:` In Lower Income Countries/Lower Middle Income Countries LIC/LMIC there is no significant difference in political agency between rural, town and city areas. In UMIC rural residents report significantly stronger perceived political agency than town (b=0.09*) and city (b=0.13**) residents - whereas political agency does not differ significantly between town and city areas. In HIC people in rural areas report significantly less political agency than people in town (b=-0.13**) and city (-0.20***) areas; there is no significant difference in the level of perceived political agency between town and city areas.

5.2 Interactions with Financial Satisfaction

Table 2 shows the average reported levels of personal and political agency across rural, town, and city areas by individuals' material resources. With regard to personal agency, people with moderate and substantial material resources follow the overall pattern, with personal agency being lowest in cities and highest in towns. Only among people with limited material resources do we observe a continuous urban-rural gradient, with perceived personal agency being highest in rural areas, followed by towns, and lowest in cities.

Similarly, the overall pattern of political agency being higher in rural areas than in towns and cities (see Table 1) is mirrored among people with moderate and substantial material resources. Only among those with limited material resources does a difference emerge between towns and cities: they report the highest levels of political agency in rural areas, followed by cities, and the lowest levels in

towns.

`Political agency` for `bottom 4` categories of `financial satisfaction`: lowest in the middle 20-500k. `Political agency` for `middle 3` categories of `financial satisfaction`: highest in the smallest places <20k; and likewise for the `top 3`.

Table 2: Means by level of financial satisfaction.

	personal agency				**political agency**			
		financial satisfaction:				financial satisfaction:		
place	overall	bottom 4	middle 3	top 3	overall	bottom 4	middle 3	top 3
<20k	7.20	6.25	7.04	8.09	2.96	2.78	2.96	3.11
20-500k	7.27	6.15	7.15	8.12	2.88	2.66	2.87	3.01
>500k	7.12	6.10	6.99	8.06	2.89	2.72	2.89	3.02

The results of the interactions with financial satisfaction for both personal and political agency are visualized in figure 2. We focus here on the visual interpretations using marginsplot, but all underlying regressions are in the appendix.

In panel (a), personal agency increases with size among respondents in the top three and also middle three financial-satisfaction categories from $< 20k$ to $> 500k$, but decreases for the bottom four categories. In other words, personal agency differences are magnified in the largest cities between financial-satisfaction groups–financially satisfied respondents report a larger advantage in freedom over financially dissatisfied respondents in cities. The corresponding interaction results for political agency in panel (b) are insignificant.

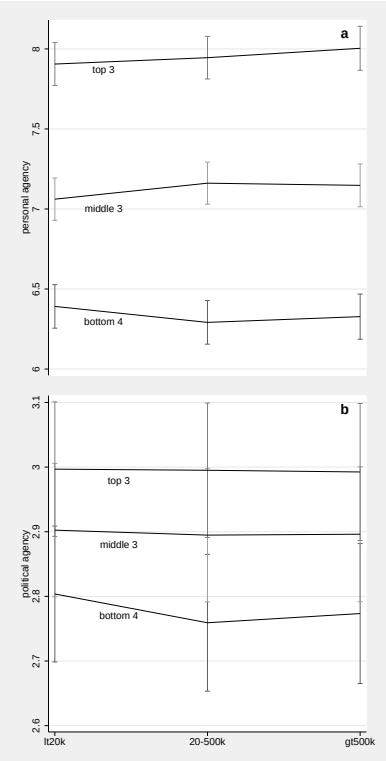


Figure 2: Predicted `personal agency` in panel a, and `political agency` in panel b on y-axes across rural-urban on x-axis by financial satisfaction. `Personal agency` adjusted for married, employment status, health, children, and religion important. `Political agency` adjusted for education, employment status, health, and religion important. Regressions are in appendix. Note: These results are robust when dropping countries where category >500k has <50obs; but not where the category <20k has <50obs.

`Personal Agency`: People with limited material resources report significantly more personal agency in rural areas than in town areas (b=0.10**); no other significant differences emerge. People with moderate material resources report more personal agency when residing in town areas (b=0.10***) and city areas (b=0.09***) as compared to rural areas, yet similar levels when comparing town and

city areas. People with substantial material resources report significantly higher levels of personal agency in city areas than rural areas (b=0.10**); no other significant differences emerge. `Political Agency:` With one exception, no significant urban-rural differences remain, once investigated separately by individuals level of material resources–suggesting that much of the urban-rural-divide in political agency is due to differences in material resources. The only small difference that emerges indicates that people with limited resources in rural areas report slightly more political agency (b=0.04*) than in town areas.

5.2.1 By Both: Country Income Group Interactions with Financial Satisfaction

In figure 3 the earlier pattern holds for LIC/LMIC and HIC, but is quite opposite in UMIC.

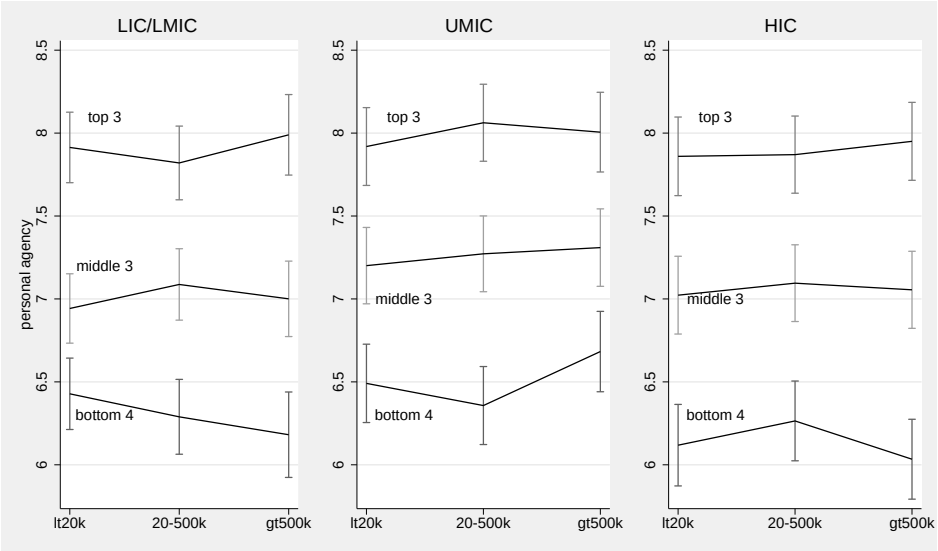


Figure 3: Predicted `personal agency` on y-axis across rural-urban on x-axis by financial satisfaction and level of development in separate panels. `Personal agency` adjusted for married, employment status, health, children, and religion important.

In figure 4 in LIC/LMIC the greatest agency decrease from rural to urban is observed for most satisfied financially; and in HIC it is all financial satisfaction groups that have more political agency in cities.

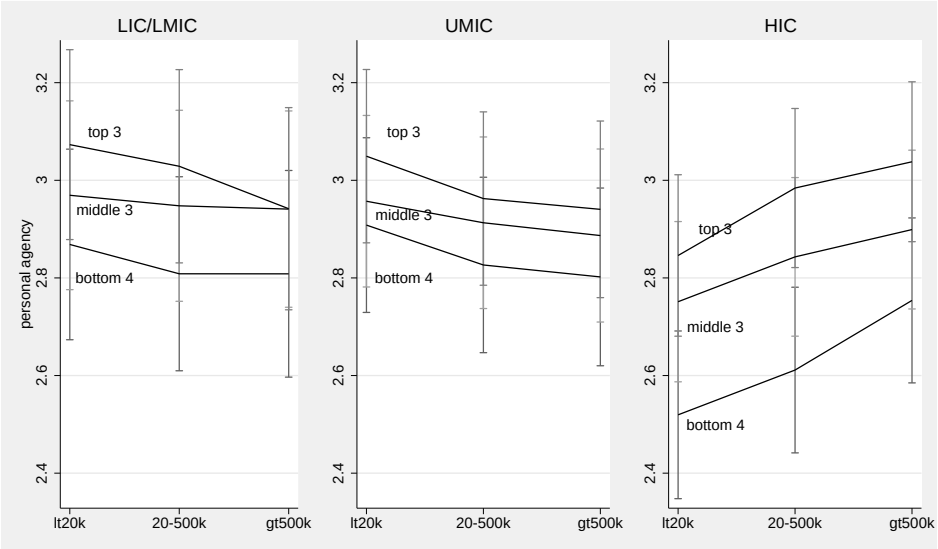


Figure 4: Predicted `political agency` on y-axis across rural-urban on x-axis by financial satisfaction and level of development in separate panels. `Political agency` adjusted for education, employment status, health, and religion important.

We also present results separately by country in the appendix.

6 Discussion and Conclusion

"In the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive." Ferdinand Toennies

The expectation that cities provide greater freedom is plausible, but in pooled models without interactions, we found no significant differences when controlling for country dummies. This is a novel contribution/study. A substantial body of research has investigated the antecedents of personal agency, yet none of these studies has focused on examination of the differences across the urban-rural gradient.

It is a novel study. One of us is an expert in agency research and literature and it is worth emphasizing that really a lot of studies investigate antecedents of personal agency–and none of them had a dedicated focus on urban/rural. It is important to underscore that even finding no difference in personal agency level between urban and rural areas is worth reporting, given our expectation of an urban advantage and a positive relationship. Contrary to the longstanding expectation from theory and literature that cities offer greater freedom and opportunities, we find little evidence of an urban advantage over rural areas in personal agency. This particular finding is important to highlight as it suggests that the greater choices and opportunities found in cities do not seem to translate into a stronger sense of control over one's life.

For political agency, however, we find it to be significantly higher in urban than in rural areas in high-income countries as hypothesized in the "left behind ruralities" hypothesis. We find support for Toennies' observation that "in the city and therefore, where general conditions characteristic of the Gesellschaft prevail, only the upper strata, the rich and the cultured, are really active and alive" (Tönnies [1887] 2002, p. 227). There are significant interactions in the pooled sample for financial satisfaction.

Consistent with the findings of Okulicz-Kozaryn (2024), urbanites probably have, in some ways, fuller and freer lives, but perhaps they realize more what they are missing relative to other urban residents, and feel relatively less free. It's possible that the availability of opportunities does not necessarily translate into the ability to use them. Costs, stress, and unequal resources may limit one's choices even when possibilities are endless. And too much freedom in general may backfire and even lead to a paralysis (Schwartz 2004).

Another mechanism could be that people tend to compare themselves with those who are better off (Frey and Stutzer 2002) and there are a lot more people to compare to in urban areas. The inequality ladder is taller and the pond is bigger–a small fish in a big pond (city) seems to be less free than a big fish in a small pond (rural area).

It is opposite to what we hypothesized: `H2:` The urban-rural divide in personal and political agency is significantly larger in low- and middle-income countries than in high-income countries. The only significant difference is actually in rich countries, where cities have more political agency than ruralities.

Overall, our findings suggest that the freedom associated with urban areas depends on which aspect we consider. Cities offer more opportunities and choices, but these do not necessarily translate into greater perceived freedom and control over one's life. Yet, for political agency, we find an urban advantage in high income countries, consistent with the "left behind ruralities" hypothesis. In high income countries, rural residents report lower political agency, even though personal agency generally shows little urban-rural difference. Hence, they may feel that they have control over their own lives while feeling that they have less say in government decisions. Future research could shed light on our findings by investigating whether rural residents perceptions of having less political voice reflect differences in institutional responsiveness, political representation, or access to decision making.

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ONLINE APPENDIX (SOM; Supp Online Mat)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead as SOM 'Supplementary Online Material']

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1 Variables' Definitions and Descriptive Statistics

Table 3: Variable definitions.

name	description
personal agency	"Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out."
political agency	"How much would you say the political system in your country allows people like you to have a say in what the government does?" Responses range from 1 () to 5 (). We reverse the response categories so that higher values indicate stronger perceived political agency.
place size	"OBSERVATIONS BY THE INTERVIEWER; Code size of town where interview was conducted"
Income Level Name	World Bank income level classification
financial satisfaction	"How satisfied are you with the financial situation of your household?"
married or living together as married	"Are you currently ....(READ OUT AND CODE ONE ONLY) 1 'Married' 2 'Living together as married' 3 'Divorced' 4 'Separated' 5 'Widowed' 6 'Single/Never married' 7 'Divorced, Separated or Widow' 8 'Living apart but steady relation (married,cohabitation)'"
Employment status	"Are you employed now or not? IF YES: About how many hours a week? If more than one job: only for the main job 1 'Full time' 2 'Part time' 3 'Self employed' 4 'Retired' 5 'Housewife' 6 'Students' 7 'Unemployed' 8 'Other'"
health	"State of health (subjective)"
religion important	"WVS2000: For each of the following aspects, indicate how important it is in your life. Would you say it is: EVS1999: Please say, for each of the following, how important it is in your life. Religion"
Education level (recoded)	Lower, Middle, Upper

Per distributions it is worth noting left-skewed distribution on personal agency, unexpectedly most people feel maximally free at 10 out of 1-10, likewise but as expected with urbanism–most people are in the top category $> 500k$, and again left-skewed on religion, probably not unexpected–religiosity is declining, but still strong.

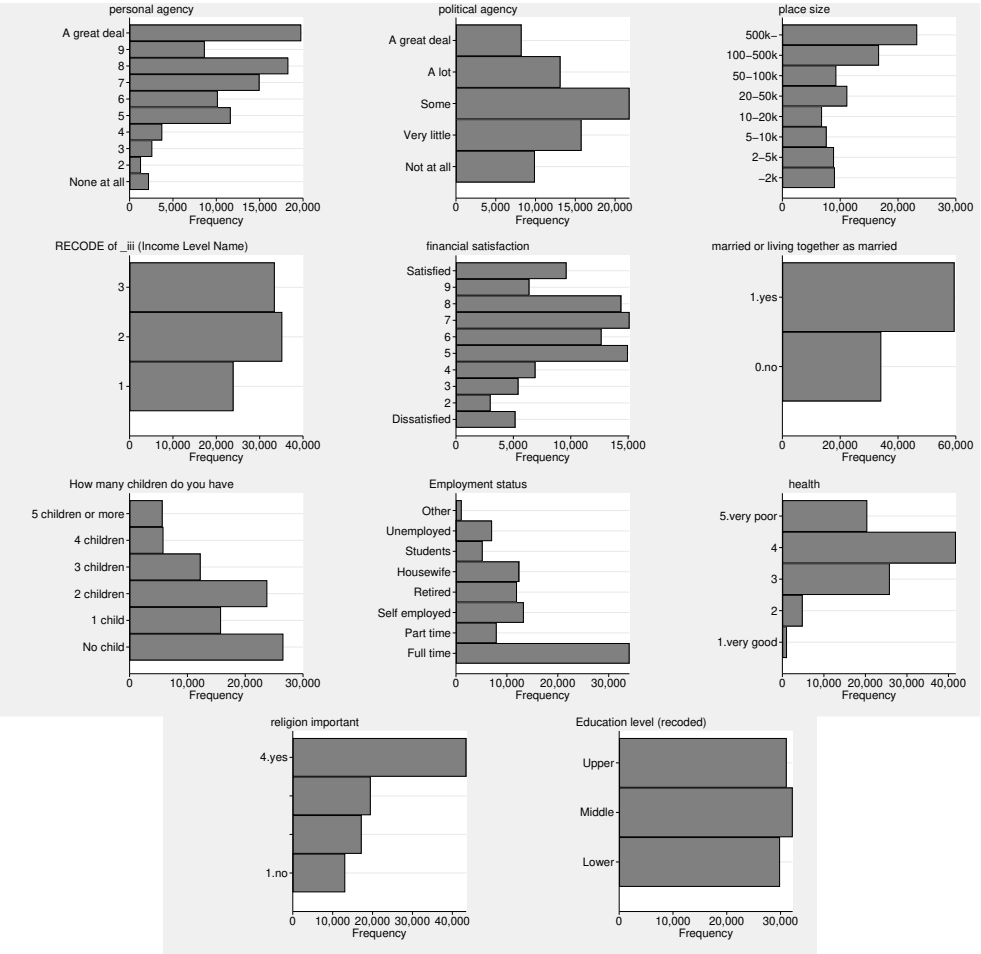


Figure 5: Variables' distribution.

2 Regressions underlying body results

```
. mixed free i.t3 i.sf3    mar i.emp health kids rel_imp   || cc:   ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -184351.37
Iteration 1:  Log likelihood = -184351.37  (backed up)
Computing standard errors ...
Mixed-effects ML regression                     Number of obs     =     85,997
Group variable: cc                              Number of groups  =         62
                                                Obs per group:
                                                              min =        849
                                                              avg =   1,387.0
                                                              max =      4,018
                                                Wald chi2(15)     =   10218.54
Log likelihood = -184351.37                     Prob > chi2       =     0.0000
------------------------------------------------------------------------------
        free | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
      20-500k |       0.03       0.02     1.69   0.09        -0.01        0.07
       gt500k |       0.05       0.02     2.34   0.02         0.01        0.10
              |
         sf3 |
           2 |       0.78       0.02    41.11   0.00         0.75        0.82
           3 |       1.61       0.02    76.03   0.00         1.56        1.65
              |
         mar |       0.02       0.02     1.28   0.20        -0.01        0.06
              |
         emp |
```

```
    Part time  |      -0.09       0.03    -3.40    0.00        -0.15       -0.04
Self employed  |      -0.01       0.02    -0.50    0.61        -0.06        0.03
      Retired  |      -0.05       0.02    -2.17    0.03        -0.10       -0.01
    Housewife  |      -0.21       0.02    -8.93    0.00        -0.26       -0.16
     Students  |      -0.21       0.03    -6.09    0.00        -0.27       -0.14
   Unemployed  |      -0.14       0.03    -4.89    0.00        -0.20       -0.09
        Other  |      -0.21       0.07    -3.10    0.00        -0.35       -0.08
               |
       health  |       0.37       0.01    42.13    0.00         0.35        0.39
         kids  |       0.03       0.01     4.47    0.00         0.01        0.04
      rel_imp  |       0.04       0.01     4.09    0.00         0.02        0.05
        _cons  |       4.81       0.08    59.68    0.00         4.65        4.97
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Identity                 |
                  var(_cons) |       0.26       0.05          0.18        0.37
-----------------------------+------------------------------------------------
               var(Residual) |       4.25       0.02          4.21        4.29
------------------------------------------------------------------------------
LR test vs. linear model: chibar2(01) = 4215.72       Prob >= chibar2 = 0.0000

. mixed polSay i.t3 i.sf3     i.X025R  i.emp health  rel_imp   || cc:   ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -103870.82
Iteration 1:  Log likelihood = -103870.82
Computing standard errors ...
Mixed-effects ML regression                         Number of obs    =  66,483
Group variable: cc                                  Number of groups =      50
                                                    Obs per group:
                                                                 min =     416
                                                                 avg = 1,329.7
                                                                 max =   3,997
                                                    Wald chi2(15)    =  860.15
Log likelihood = -103870.82                         Prob > chi2      =  0.0000
------------------------------------------------------------------------------
       polSay | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
--------------+---------------------------------------------------------------
           t3 |
     20-500k  |      -0.01       0.01    -1.24    0.21        -0.04        0.01
      gt500k  |      -0.01       0.01    -0.81    0.42        -0.04        0.02
              |
          sf3 |
            2 |       0.12       0.01     9.64    0.00         0.09        0.14
            3 |       0.21       0.01    16.03    0.00         0.19        0.24
              |
        X025R |
       Middle |       0.00       0.01     0.02    0.99        -0.02        0.02
        Upper |       0.04       0.01     3.43    0.00         0.02        0.07
              |
          emp |
    Part time |      -0.00       0.02    -0.18    0.86        -0.04        0.03
Self employed |      -0.02       0.01    -1.38    0.17        -0.05        0.01
      Retired |       0.00       0.02     0.19    0.85        -0.03        0.03
    Housewife |      -0.07       0.02    -4.37    0.00        -0.10       -0.04
     Students |       0.00       0.02     0.24    0.81        -0.04        0.05
   Unemployed |      -0.06       0.02    -3.24    0.00        -0.10       -0.02
        Other |      -0.02       0.05    -0.50    0.62        -0.11        0.07
              |
       health |       0.08       0.01    13.80    0.00         0.06        0.09
      rel_imp |       0.07       0.01    12.79    0.00         0.06        0.08
        _cons |       2.28       0.06    38.35    0.00         2.17        2.40
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Identity                 |
                  var(_cons) |       0.13       0.03          0.09        0.20
-----------------------------+------------------------------------------------
               var(Residual) |       1.33       0.01          1.31        1.34
------------------------------------------------------------------------------
LR test vs. linear model: chibar2(01) = 5173.88       Prob >= chibar2 = 0.0000
.
```


2.1 Interactions with level of development

```
 mixed polSay i.t3##i.iii  i.X025R i.emp satFin health  rel_imp  || cc: i.t3  ,mle //final! yes educ; no mar no kids
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -101253.19
Iteration 1:  Log likelihood = -101253.19
Computing standard errors ...
Mixed-effects ML regression                         Number of obs    =  64,862
Group variable: cc                                  Number of groups =      48
                                                    Obs per group:
                                                                 min =     726
                                                                 avg = 1,351.3
                                                                 max =   3,997
                                                    Wald chi2(20)    =  890.11
Log likelihood = -101253.19                         Prob > chi2      =  0.0000
------------------------------------------------------------------------------
       polSay | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
--------------+---------------------------------------------------------------
           t3 |
     20-500k  |      -0.04       0.04    -0.96    0.34        -0.11        0.04
      gt500k  |      -0.06       0.06    -1.07    0.28        -0.18        0.05
              |
          iii |
```

```
           2  |      -0.01       0.13    -0.09    0.93        -0.26        0.24
           3  |      -0.20       0.14    -1.49    0.14        -0.47        0.06
              |
       t3#iii |
    20-500k#2  |      -0.05       0.05    -1.02    0.31        -0.16        0.05
    20-500k#3  |       0.16       0.06     2.73    0.01         0.05        0.28
     gt500k#2  |      -0.07       0.08    -0.87    0.39        -0.22        0.09
     gt500k#3  |       0.26       0.09     2.87    0.00         0.08        0.44
              |
         X025R |
       Middle  |       0.01       0.01     0.50    0.62        -0.02        0.03
        Upper  |       0.05       0.01     3.78    0.00         0.02        0.07
              |
           emp |
    Part time  |       0.00       0.02     0.04    0.97        -0.03        0.03
Self employed  |      -0.02       0.01    -1.58    0.11        -0.05        0.01
      Retired  |       0.01       0.02     0.37    0.71        -0.03        0.04
    Housewife  |      -0.07       0.02    -4.40    0.00        -0.10       -0.04
     Students  |       0.00       0.02     0.19    0.85        -0.04        0.04
   Unemployed  |      -0.06       0.02    -3.09    0.00        -0.10       -0.02
        Other  |      -0.03       0.05    -0.72    0.47        -0.13        0.06
              |
       satFin  |       0.03       0.00    17.07    0.00         0.03        0.04
       health  |       0.07       0.01    12.53    0.00         0.06        0.08
      rel_imp  |       0.07       0.01    12.29    0.00         0.06        0.08
        _cons  |       2.27       0.09    23.94    0.00         2.09        2.46
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Independent              |
                   var(2.t3) |       0.02       0.00          0.01        0.03
                   var(3.t3) |       0.03       0.01          0.02        0.06
                  var(_cons) |       0.13       0.03          0.09        0.20
-----------------------------+------------------------------------------------
               var(Residual) |       1.32       0.01          1.31        1.34
------------------------------------------------------------------------------
LR test vs. linear model: chi2(3) = 4821.97               Prob > chi2 = 0.0000
Note: LR test is conservative and provided only for reference.
.
. margins,  at(t3=(1 2 3)  iii=(1 2 3))
Predictive margins                                      Number of obs = 64,862
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       iii = 1
2._at: t3  = 1
       iii = 2
3._at: t3  = 1
       iii = 3
4._at: t3  = 2
       iii = 1
5._at: t3  = 2
       iii = 2
6._at: t3  = 2
       iii = 3
7._at: t3  = 3
       iii = 1
8._at: t3  = 3
       iii = 2
9._at: t3  = 3
       iii = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       2.97       0.09    33.11    0.00         2.79        3.14
          2  |       2.96       0.09    32.86    0.00         2.78        3.13
          3  |       2.77       0.10    26.97    0.00         2.56        2.97
          4  |       2.93       0.10    30.80    0.00         2.75        3.12
          5  |       2.87       0.09    30.20    0.00         2.68        3.05
          6  |       2.89       0.11    26.58    0.00         2.68        3.11
          7  |       2.90       0.11    27.41    0.00         2.70        3.11
          8  |       2.82       0.10    28.10    0.00         2.63        3.02
          9  |       2.96       0.11    26.19    0.00         2.74        3.19
------------------------------------------------------------------------------
. marginsplot,saving(a1,replace)ytitle("political say")scheme(s2mono)title("") text(2.98 1.4 "low and lower middle")text(2.9 1.4 "uper middle")
> text(2.78 1.4 "high")recast(line) legend(off)xtitle("size of a place") //xsc(off)
Variables that uniquely identify margins: t3 iii
file a1.gph saved

. mixed free i.t3##i.iii  mar i.emp satFin health kids rel_imp  || cc:  i.t3 ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -181550.65
Iteration 1:  Log likelihood = -181550.65
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =   84,775
Group variable: cc                              Number of groups =       61
                                                Obs per group:
                                                              min =      849
                                                              avg =  1,389.8
                                                              max =    4,018
                                                Wald chi2(20)    = 10564.59
Log likelihood = -181550.65                     Prob > chi2      =   0.0000
------------------------------------------------------------------------------
        free | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
    20-500k  |       0.01       0.06     0.17    0.86        -0.10        0.12
     gt500k  |       0.09       0.08     1.04    0.30        -0.08        0.25
             |
         iii |
          2  |       0.12       0.17     0.74    0.46        -0.20        0.45
          3  |      -0.03       0.18    -0.16    0.87        -0.37        0.32
             |
       t3#iii |
  20-500k#2  |       0.03       0.07     0.40    0.69        -0.12        0.18
```

```
    20-500k#3  |       0.05       0.08     0.67    0.50        -0.10        0.21
     gt500k#2  |       0.04       0.11     0.36    0.72        -0.17        0.24
     gt500k#3  |      -0.12       0.12    -1.00    0.32        -0.35        0.11
               |
           mar |       0.02       0.02     1.38    0.17        -0.01        0.06
               |
           emp |
    Part time  |      -0.09       0.03    -3.31    0.00        -0.15       -0.04
Self employed  |      -0.01       0.02    -0.44    0.66        -0.06        0.04
      Retired  |      -0.06       0.02    -2.33    0.02        -0.11       -0.01
    Housewife  |      -0.20       0.02    -8.49    0.00        -0.25       -0.16
     Students  |      -0.21       0.03    -6.12    0.00        -0.27       -0.14
   Unemployed  |      -0.11       0.03    -3.71    0.00        -0.17       -0.05
        Other  |      -0.17       0.07    -2.49    0.01        -0.31       -0.04
               |
       satFin  |       0.26       0.00    79.88    0.00         0.25        0.26
       health  |       0.34       0.01    38.79    0.00         0.33        0.36
         kids  |       0.03       0.01     4.58    0.00         0.01        0.04
      rel_imp  |       0.03       0.01     3.81    0.00         0.02        0.05
        _cons  |       4.16       0.13    30.94    0.00         3.90        4.42
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.      [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Independent              |
                   var(2.t3) |       0.03       0.01          0.02        0.06
                   var(3.t3) |       0.06       0.02          0.03        0.11
                  var(_cons) |       0.27       0.05          0.19        0.39
-----------------------------+------------------------------------------------
               var(Residual) |       4.22       0.02          4.18        4.26
------------------------------------------------------------------------------
LR test vs. linear model: chi2(3) = 4153.75                   Prob > chi2 = 0.0000
Note: LR test is conservative and provided only for reference.
. margins,  at(t3=(1 2 3)  iii=(1 2 3))
Predictive margins                                      Number of obs = 84,775
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       iii = 1
2._at: t3  = 1
       iii = 2
3._at: t3  = 1
       iii = 3
4._at: t3  = 2
       iii = 1
5._at: t3  = 2
       iii = 2
6._at: t3  = 2
       iii = 3
7._at: t3  = 3
       iii = 1
8._at: t3  = 3
       iii = 2
9._at: t3  = 3
       iii = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       7.15       0.13    57.16    0.00         6.90        7.40
          2  |       7.27       0.11    65.26    0.00         7.06        7.49
          3  |       7.12       0.12    57.64    0.00         6.88        7.36
          4  |       7.16       0.13    53.85    0.00         6.90        7.42
          5  |       7.31       0.12    62.48    0.00         7.08        7.54
          6  |       7.19       0.13    55.87    0.00         6.93        7.44
          7  |       7.24       0.15    49.35    0.00         6.95        7.52
          8  |       7.40       0.12    59.72    0.00         7.16        7.64
          9  |       7.09       0.13    52.58    0.00         6.83        7.35
------------------------------------------------------------------------------
. marginsplot,saving(a2,replace)ytitle("freedom")scheme(s2mono)title("") text(7.25 2.7 "low and lower middle")text(7.33 1.4 "uper middle")text(
> 7.1 2.7 "high")recast(line) legend(off)xtitle("size of a place") xsc(off)
Variables that uniquely identify margins: t3 iii
file a2.gph saved
graph combine a2.gph a1.gph, cols(1)  xsize(2) ysize(4) scale(1.1)
gr export m-fps.pdf,replace

    .19
```

2.1.1 which differences are significant at what level

`Personal Agency:` With one exception, there are no significant differences in the level of perceived agency between rural, town, and city areas irrespective of the level of development: In UMIC, people in city areas report significantly more personal agency than in rural areas (b=0.12*).

`Political Agency:` In LIC/LMIC there is no significant difference in political agency between rural, town and city areas. In UMIC rural residents report significantly stronger perceived political agency than town (b=0.09*) and city (b=0.13**) residents - whereas political agency does not differ significantly between town and city areas. In HIC people in rural areas report significantly less political agency than people in town (b=-0.13**) and city (-0.20***) areas; there is no significant difference in the level of perceived political agency between town and city areas.

```
//this is for free
FIRST JUST ONLY RELEVANT ONES, and below that all
```

	Contrast	Delta-method std. err.	z	Unadjusted P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	0.01	0.06	0.17	0.86	-0.10	0.12
(1#gt500k) vs (1#lt20k)	0.09	0.08	1.04	0.30	-0.08	0.25
(1#gt500k) vs (1#20-500k)	0.08	0.09	0.82	0.41	-0.11	0.26
(2#20-500k) vs (2#lt20k)	0.04	0.05	0.80	0.42	-0.06	0.14
(2#gt500k) vs (2#lt20k)	0.12	0.06	1.96	0.05	-0.00	0.25
(2#gt500k) vs (2#20-500k)	0.08	0.07	1.16	0.25	-0.06	0.23
(3#20-500k) vs (3#lt20k)	0.06	0.06	1.08	0.28	-0.05	0.18
(3#gt500k) vs (3#lt20k)	-0.03	0.08	-0.38	0.71	-0.19	0.13
(3#gt500k) vs (3#20-500k)	-0.10	0.09	-1.08	0.28	-0.27	0.08

//this is for free

```
. margins t3, at(iii=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins              Number of obs = 84,775
Expression: Linear prediction, fixed portion, predict()
1._at: iii = 1
2._at: iii = 2
3._at: iii = 3
```

	Contrast	Delta-method std. err.	z	Unadjusted P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	0.01	0.06	0.17	0.86	-0.10	0.12
(1#gt500k) vs (1#lt20k)	0.09	0.08	1.04	0.30	-0.08	0.25
(2#lt20k) vs (1#lt20k)	0.12	0.17	0.74	0.46	-0.20	0.45
(2#20-500k) vs (1#lt20k)	0.16	0.17	0.96	0.34	-0.17	0.50
(2#gt500k) vs (1#lt20k)	0.25	0.18	1.41	0.16	-0.10	0.59
(3#lt20k) vs (1#lt20k)	-0.03	0.18	-0.16	0.87	-0.37	0.32
(3#20-500k) vs (1#lt20k)	0.04	0.18	0.20	0.84	-0.32	0.39
(3#gt500k) vs (1#lt20k)	-0.06	0.18	-0.33	0.75	-0.42	0.30
(1#gt500k) vs (1#20-500k)	0.08	0.09	0.82	0.41	-0.11	0.26
(2#lt20k) vs (1#20-500k)	0.11	0.17	0.66	0.51	-0.23	0.45
(2#20-500k) vs (1#20-500k)	0.15	0.18	0.87	0.38	-0.19	0.50
(2#gt500k) vs (1#20-500k)	0.24	0.18	1.32	0.19	-0.12	0.60
(3#lt20k) vs (1#20-500k)	-0.04	0.18	-0.21	0.83	-0.39	0.32
(3#20-500k) vs (1#20-500k)	0.03	0.19	0.14	0.89	-0.34	0.39
(3#gt500k) vs (1#20-500k)	-0.07	0.19	-0.37	0.71	-0.44	0.30
(2#lt20k) vs (1#gt500k)	0.04	0.18	0.20	0.84	-0.32	0.40
(2#20-500k) vs (1#gt500k)	0.08	0.19	0.41	0.68	-0.29	0.44
(2#gt500k) vs (1#gt500k)	0.16	0.19	0.84	0.40	-0.21	0.54
(3#lt20k) vs (1#gt500k)	-0.12	0.19	-0.60	0.55	-0.49	0.26
(3#20-500k) vs (1#gt500k)	-0.05	0.20	-0.26	0.79	-0.43	0.33
(3#gt500k) vs (1#gt500k)	-0.15	0.20	-0.74	0.46	-0.54	0.24
(2#20-500k) vs (2#lt20k)	0.04	0.05	0.80	0.42	-0.06	0.14
(2#gt500k) vs (2#lt20k)	0.12	0.06	1.96	0.05	-0.00	0.25
(3#lt20k) vs (2#lt20k)	-0.15	0.17	-0.92	0.36	-0.48	0.17
(3#20-500k) vs (2#lt20k)	-0.09	0.17	-0.52	0.60	-0.42	0.25
(3#gt500k) vs (2#lt20k)	-0.18	0.18	-1.05	0.29	-0.53	0.16
(2#gt500k) vs (2#20-500k)	0.08	0.07	1.16	0.25	-0.06	0.23
(3#lt20k) vs (2#20-500k)	-0.19	0.17	-1.13	0.26	-0.53	0.14
(3#20-500k) vs (2#20-500k)	-0.13	0.17	-0.74	0.46	-0.47	0.21
(3#gt500k) vs (2#20-500k)	-0.22	0.18	-1.25	0.21	-0.57	0.13
(3#lt20k) vs (2#gt500k)	-0.28	0.18	-1.59	0.11	-0.62	0.07
(3#20-500k) vs (2#gt500k)	-0.21	0.18	-1.19	0.23	-0.56	0.14
(3#gt500k) vs (2#gt500k)	-0.31	0.18	-1.68	0.09	-0.67	0.05
(3#20-500k) vs (3#lt20k)	0.06	0.06	1.08	0.28	-0.05	0.18
(3#gt500k) vs (3#lt20k)	-0.03	0.08	-0.38	0.71	-0.19	0.13
(3#gt500k) vs (3#20-500k)	-0.10	0.09	-1.08	0.28	-0.27	0.08

//this is for PolSay
FIRST JUST ONLY RELEVANT ONES, and below that all

	Contrast	Delta-method std. err.	z	Unadjusted P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	-0.04	0.04	-0.96	0.34	-0.11	0.04
(1#gt500k) vs (1#lt20k)	-0.06	0.06	-1.07	0.28	-0.18	0.05
(1#gt500k) vs (1#20-500k)	-0.03	0.07	-0.42	0.67	-0.16	0.10
(2#20-500k) vs (2#lt20k)	-0.09	0.04	-2.41	0.02	-0.16	-0.02
(2#gt500k) vs (2#lt20k)	-0.13	0.05	-2.63	0.01	-0.23	-0.03
(2#gt500k) vs (2#20-500k)	-0.04	0.06	-0.72	0.47	-0.16	0.07
(3#20-500k) vs (3#lt20k)	0.13	0.05	2.74	0.01	0.04	0.22
(3#gt500k) vs (3#lt20k)	0.20	0.07	2.88	0.00	0.06	0.33
(3#gt500k) vs (3#20-500k)	0.07	0.08	0.93	0.35	-0.08	0.22

//this is for PolSay

```
 margins t3, at(iii=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins              Number of obs = 64,862
Expression: Linear prediction, fixed portion, predict()
1._at: iii = 1
2._at: iii = 2
3._at: iii = 3
```

	Contrast	Delta-method std. err.	z	Unadjusted P>|z|	Unadjusted [95% conf.	interval]
_at#t3						
(1#20-500k) vs (1#lt20k)	-0.04	0.04	-0.96	0.34	-0.11	0.04
(1#gt500k) vs (1#lt20k)	-0.06	0.06	-1.07	0.28	-0.18	0.05
(2#lt20k) vs (1#lt20k)	-0.01	0.13	-0.09	0.93	-0.26	0.24
(2#20-500k) vs (1#lt20k)	-0.10	0.13	-0.78	0.43	-0.36	0.15
(2#gt500k) vs (1#lt20k)	-0.14	0.13	-1.07	0.28	-0.41	0.12
(3#lt20k) vs (1#lt20k)	-0.20	0.14	-1.49	0.14	-0.47	0.06
(3#20-500k) vs (1#lt20k)	-0.08	0.14	-0.55	0.59	-0.35	0.20

```
   (3#gt500k) vs (1#lt20k)  |      -0.01       0.14    -0.03    0.97        -0.29        0.28
 (1#gt500k) vs (1#20-500k)  |      -0.03       0.07    -0.42    0.67        -0.16        0.10
  (2#lt20k) vs (1#20-500k)  |       0.02       0.13     0.19    0.85        -0.23        0.28
(2#20-500k) vs (1#20-500k)  |      -0.07       0.13    -0.49    0.62        -0.33        0.20
 (2#gt500k) vs (1#20-500k)  |      -0.11       0.14    -0.78    0.43        -0.38        0.16
  (3#lt20k) vs (1#20-500k)  |      -0.17       0.14    -1.19    0.23        -0.44        0.11
(3#20-500k) vs (1#20-500k)  |      -0.04       0.14    -0.28    0.78        -0.32        0.24
 (3#gt500k) vs (1#20-500k)  |       0.03       0.15     0.21    0.83        -0.26        0.32
   (2#lt20k) vs (1#gt500k)  |       0.05       0.14     0.38    0.70        -0.22        0.33
 (2#20-500k) vs (1#gt500k)  |      -0.04       0.14    -0.26    0.79        -0.32        0.24
  (2#gt500k) vs (1#gt500k)  |      -0.08       0.15    -0.55    0.58        -0.37        0.21
   (3#lt20k) vs (1#gt500k)  |      -0.14       0.15    -0.94    0.35        -0.43        0.15
 (3#20-500k) vs (1#gt500k)  |      -0.01       0.15    -0.08    0.93        -0.31        0.29
  (3#gt500k) vs (1#gt500k)  |       0.06       0.16     0.38    0.70        -0.24        0.36
 (2#20-500k) vs (2#lt20k)   |      -0.09       0.04    -2.41    0.02        -0.16       -0.02
  (2#gt500k) vs (2#lt20k)   |      -0.13       0.05    -2.63    0.01        -0.23       -0.03
   (3#lt20k) vs (2#lt20k)   |      -0.19       0.14    -1.40    0.16        -0.46        0.08
 (3#20-500k) vs (2#lt20k)   |      -0.07       0.14    -0.46    0.64        -0.34        0.21
  (3#gt500k) vs (2#lt20k)   |       0.01       0.14     0.05    0.96        -0.28        0.29
 (2#gt500k) vs (2#20-500k)  |      -0.04       0.06    -0.72    0.47        -0.16        0.07
  (3#lt20k) vs (2#20-500k)  |      -0.10       0.14    -0.73    0.47        -0.38        0.17
(3#20-500k) vs (2#20-500k)  |       0.02       0.14     0.17    0.86        -0.26        0.31
 (3#gt500k) vs (2#20-500k)  |       0.10       0.15     0.66    0.51        -0.19        0.39
   (3#lt20k) vs (2#gt500k)  |      -0.06       0.14    -0.41    0.68        -0.34        0.22
 (3#20-500k) vs (2#gt500k)  |       0.07       0.15     0.45    0.65        -0.22        0.36
  (3#gt500k) vs (2#gt500k)  |       0.14       0.15     0.92    0.36        -0.16        0.44
 (3#20-500k) vs (3#lt20k)   |       0.13       0.05     2.74    0.01         0.04        0.22
  (3#gt500k) vs (3#lt20k)   |       0.20       0.07     2.88    0.00         0.06        0.33
 (3#gt500k) vs (3#20-500k)  |       0.07       0.08     0.93    0.35        -0.08        0.22
-----------------------------------------------------------------------------------------------
```

.19

2.2 Interactions with Financial Satisfaction

```
. mixed free i.t3##i.sf3    mar i.emp health kids rel_imp  || cc:   ,mle
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -184338.27
Iteration 1:  Log likelihood = -184338.27
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =   85,997
Group variable: cc                              Number of groups =       62
                                                Obs per group:
                                                              min =      849
                                                              avg =  1,387.0
                                                              max =    4,018
                                                Wald chi2(19)    = 10247.87
Log likelihood = -184338.27                     Prob > chi2      =   0.0000
------------------------------------------------------------------------------
        free | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
    20-500k  |      -0.10       0.04    -2.80    0.01        -0.17       -0.03
     gt500k  |      -0.06       0.04    -1.56    0.12        -0.14        0.02
             |
         sf3 |
           2 |       0.67       0.03    21.86    0.00         0.61        0.73
           3 |       1.51       0.03    45.85    0.00         1.45        1.58
             |
       t3#sf3 |
  20-500k#2  |       0.20       0.04     4.69    0.00         0.12        0.28
  20-500k#3  |       0.14       0.05     3.08    0.00         0.05        0.23
   gt500k#2  |       0.15       0.05     3.15    0.00         0.06        0.24
   gt500k#3  |       0.16       0.05     3.16    0.00         0.06        0.26
             |
         mar |       0.02       0.02     1.24    0.21        -0.01        0.05
             |
         emp |
  Part time  |      -0.09       0.03    -3.38    0.00        -0.15       -0.04
Self employed |      -0.01       0.02    -0.50    0.62        -0.06        0.03
     Retired |      -0.05       0.02    -2.17    0.03        -0.10       -0.01
   Housewife |      -0.21       0.02    -8.89    0.00        -0.26       -0.16
    Students |      -0.21       0.03    -6.10    0.00        -0.27       -0.14
  Unemployed |      -0.14       0.03    -4.83    0.00        -0.20       -0.08
       Other |      -0.21       0.07    -3.06    0.00        -0.35       -0.08
             |
      health |       0.37       0.01    42.11    0.00         0.35        0.39
        kids |       0.03       0.01     4.45    0.00         0.01        0.04
     rel_imp |       0.04       0.01     4.11    0.00         0.02        0.05
       _cons |       4.89       0.08    59.41    0.00         4.73        5.05
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Identity                 |
                  var(_cons) |       0.26       0.05          0.18        0.37
-----------------------------+------------------------------------------------
               var(Residual) |       4.25       0.02          4.21        4.29
------------------------------------------------------------------------------
LR test vs. linear model: chibar2(01) = 4212.60       Prob >= chibar2 = 0.0000
. est sto a
. margins,  at(t3=(1 2 3)  sf=(1 2 3))
Predictive margins                                  Number of obs = 85,997
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       sf3 = 1
2._at: t3  = 1
       sf3 = 2
3._at: t3  = 1
```

```
       sf3 = 3
4._at: t3  = 2
       sf3 = 1
5._at: t3  = 2
       sf3 = 2
6._at: t3  = 2
       sf3 = 3
7._at: t3  = 3
       sf3 = 1
8._at: t3  = 3
       sf3 = 2
9._at: t3  = 3
       sf3 = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       6.39       0.07    92.12    0.00         6.26        6.53
          2  |       7.06       0.07   104.88    0.00         6.93        7.19
          3  |       7.91       0.07   115.69    0.00         7.77        8.04
          4  |       6.29       0.07    90.53    0.00         6.16        6.43
          5  |       7.16       0.07   106.88    0.00         7.03        7.29
          6  |       7.95       0.07   116.96    0.00         7.81        8.08
          7  |       6.33       0.07    87.90    0.00         6.19        6.47
          8  |       7.15       0.07   104.65    0.00         7.01        7.28
          9  |       8.00       0.07   114.23    0.00         7.87        8.14
------------------------------------------------------------------------------
. marginsplot, xdimension(t3) xsize(2) ysize(2) saving(z1,replace) text(8.25 4 "{bf:a}", size(large))legend(off)text(7.87 1.32 "top 3 cat")text
> (7 1.32 "middle 3 cat")text(6.3 1.32 "bottom 4 cat")title("")ytitle("personal agency")scheme(s2mono)xsc(off)recast(line) //xtitle("rural-urba
> n")
Variables that uniquely identify margins: t3 sf3
file z1.gph saved
. mixed polSay i.t3##i.sf3   i.X025R  i.emp health  rel_imp  || cc:    ,mle //final! ditched mar and kids and added educ
Performing EM optimization ...
Performing gradient-based optimization:
Iteration 0:  Log likelihood = -103869.52
Iteration 1:  Log likelihood = -103869.52  (backed up)
Computing standard errors ...
Mixed-effects ML regression                     Number of obs    =  66,483
Group variable: cc                              Number of groups =      50
                                                Obs per group:
                                                              min =     416
                                                              avg = 1,329.7
                                                              max =   3,997
                                                Wald chi2(19)    =  862.78
Log likelihood = -103869.52                     Prob > chi2      =  0.0000
------------------------------------------------------------------------------
      polSay | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
          t3 |
      20-500k |      -0.04       0.02    -1.99    0.05        -0.09       -0.00
       gt500k |      -0.03       0.03    -1.17    0.24        -0.08        0.02
              |
         sf3 |
           2 |       0.10       0.02     5.22    0.00         0.06        0.14
           3 |       0.19       0.02     9.56    0.00         0.15        0.23
              |
       t3#sf3 |
    20-500k#2 |       0.04       0.03     1.35    0.18        -0.02        0.09
    20-500k#3 |       0.04       0.03     1.50    0.13        -0.01        0.10
     gt500k#2 |       0.02       0.03     0.80    0.42        -0.03        0.08
     gt500k#3 |       0.03       0.03     0.81    0.42        -0.04        0.09
              |
        X025R |
       Middle |       0.00       0.01     0.03    0.98        -0.02        0.02
        Upper |       0.04       0.01     3.42    0.00         0.02        0.07
              |
          emp |
    Part time |      -0.00       0.02    -0.18    0.86        -0.04        0.03
Self employed |      -0.02       0.01    -1.38    0.17        -0.05        0.01
      Retired |       0.00       0.02     0.17    0.86        -0.03        0.03
    Housewife |      -0.07       0.02    -4.36    0.00        -0.10       -0.04
     Students |       0.00       0.02     0.24    0.81        -0.04        0.05
   Unemployed |      -0.06       0.02    -3.22    0.00        -0.10       -0.02
        Other |      -0.02       0.05    -0.49    0.62        -0.11        0.07
              |
       health |       0.08       0.01    13.80    0.00         0.06        0.09
      rel_imp |       0.07       0.01    12.79    0.00         0.06        0.08
        _cons |       2.30       0.06    38.08    0.00         2.18        2.42
------------------------------------------------------------------------------
------------------------------------------------------------------------------
  Random-effects parameters  |   Estimate   Std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
cc: Identity                 |
                  var(_cons) |       0.13       0.03          0.09        0.20
-----------------------------+------------------------------------------------
               var(Residual) |       1.33       0.01          1.31        1.34
------------------------------------------------------------------------------
LR test vs. linear model: chibar2(01) = 5171.45       Prob >= chibar2 = 0.0000
. est sto a
. margins,  at(t3=(1 2 3)  sf=(1 2 3))
Predictive margins                                  Number of obs = 66,483
Expression: Linear prediction, fixed portion, predict()
1._at: t3  = 1
       sf3 = 1
2._at: t3  = 1
       sf3 = 2
3._at: t3  = 1
       sf3 = 3
4._at: t3  = 2
       sf3 = 1
5._at: t3  = 2
       sf3 = 2
6._at: t3  = 2
       sf3 = 3
7._at: t3  = 3
       sf3 = 1
```

```
8._at: t3  = 3
       sf3 = 2
9._at: t3  = 3
       sf3 = 3
------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |       2.80       0.05    52.26    0.00         2.70        2.91
          2  |       2.90       0.05    55.09    0.00         2.80        3.01
          3  |       3.00       0.05    56.44    0.00         2.89        3.10
          4  |       2.76       0.05    51.09    0.00         2.65        2.87
          5  |       2.89       0.05    54.94    0.00         2.79        3.00
          6  |       3.00       0.05    56.43    0.00         2.89        3.10
          7  |       2.77       0.06    50.20    0.00         2.67        2.88
          8  |       2.90       0.05    54.34    0.00         2.79        3.00
          9  |       2.99       0.05    55.34    0.00         2.89        3.10
------------------------------------------------------------------------------
. marginsplot ,xdimension(t3) xsize(2) ysize(2) saving(z2,replace) text(3.12 4 "{bf:b}", size(large))legend(off)text(2.98 1.42 "top 3 cat")text
> (2.88 1.42 "middle 3 cat")text(2.77 1.42 "bottom 4 cat")title("")ytitle("political efficacy")xtitle("rural-urban")scheme(s2mono)recast(line)
Variables that uniquely identify margins: t3 sf3
file z2.gph saved
. graph combine z1.gph z2.gph, cols(1) scheme(s2mono) xsize(2) ysize(4) scale(1)
. gr export zz.pdf,replace
file zz.pdf saved as PDF format
```


2.2.1 which differences are significant at what level

`Personal Agency`: People with limited material resources report significantly more personal agency in rural areas than in town areas (b=0.10**); no other significant differences emerge. People with moderate material resources report more personal agency when residing in town areas (b=0.10***) and city areas (b=0.09***) as compared to rural areas, yet similar levels when comparing town and city areas. People with substantial material resources report significantly higher levels of personal agency in city areas than rural areas (b=0.10**); no other significant differences emerge.

`Political Agency`: With one exception, no significant urban-rural differences remain, once investigated separately by individuals level of material resources - suggesting that much of the urban-rural-divide in political agency is due to differences in material resources. The only small difference that emerges indicates that people with limited resources in rural areas report slightly more political agency (b=0.04*) than in town areas

```
. //this is for free
FIRST JUST ONLY RELEVANT ONES, and below that all
------------------------------------------------------------------------------------------
                             |            Delta-method    Unadjusted           Unadjusted
                             |   Contrast   std. err.      z    P>|z|     [95% conf. interval]
-----------------------------+------------------------------------------------------------
                      _at#t3 |
  (1#20-500k) vs (1#lt20k)   |      -0.10       0.04    -2.80    0.01        -0.17       -0.03
   (1#gt500k) vs (1#lt20k)   |      -0.06       0.04    -1.56    0.12        -0.14        0.02
 (1#gt500k) vs (1#20-500k)   |       0.04       0.04     0.88    0.38        -0.04        0.12
  (2#20-500k) vs (2#lt20k)   |       0.10       0.03     3.77    0.00         0.05        0.15
   (2#gt500k) vs (2#lt20k)   |       0.09       0.03     2.85    0.00         0.03        0.15
 (2#gt500k) vs (2#20-500k)   |      -0.01       0.03    -0.47    0.64        -0.07        0.04
  (3#20-500k) vs (3#lt20k)   |       0.04       0.03     1.29    0.20        -0.02        0.10
   (3#gt500k) vs (3#lt20k)   |       0.10       0.04     2.77    0.01         0.03        0.17
 (3#gt500k) vs (3#20-500k)   |       0.06       0.03     1.71    0.09        -0.01        0.13
------------------------------------------------------------------------------------------
. //this is for free
. margins t3, at(sf3=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins             Number of obs = 85,997
Expression: Linear prediction, fixed portion, predict()
1._at: sf3 = 1
2._at: sf3 = 2
3._at: sf3 = 3
------------------------------------------------------------------------------------------
                             |            Delta-method    Unadjusted           Unadjusted
                             |   Contrast   std. err.      z    P>|z|     [95% conf. interval]
-----------------------------+------------------------------------------------------------
                      _at#t3 |
  (1#20-500k) vs (1#lt20k)   |      -0.10       0.04    -2.80    0.01        -0.17       -0.03
   (1#gt500k) vs (1#lt20k)   |      -0.06       0.04    -1.56    0.12        -0.14        0.02
    (2#lt20k) vs (1#lt20k)   |       0.67       0.03    21.86    0.00         0.61        0.73
  (2#20-500k) vs (1#lt20k)   |       0.77       0.03    24.51    0.00         0.71        0.83
   (2#gt500k) vs (1#lt20k)   |       0.76       0.03    21.82    0.00         0.69        0.82
    (3#lt20k) vs (1#lt20k)   |       1.51       0.03    45.85    0.00         1.45        1.58
  (3#20-500k) vs (1#lt20k)   |       1.55       0.03    46.14    0.00         1.49        1.62
   (3#gt500k) vs (1#lt20k)   |       1.61       0.04    42.28    0.00         1.54        1.69
 (1#gt500k) vs (1#20-500k)   |       0.04       0.04     0.88    0.38        -0.04        0.12
  (2#lt20k) vs (1#20-500k)   |       0.77       0.03    23.92    0.00         0.71        0.83
(2#20-500k) vs (1#20-500k)   |       0.87       0.03    28.57    0.00         0.81        0.93
 (2#gt500k) vs (1#20-500k)   |       0.86       0.03    24.99    0.00         0.79        0.92
  (3#lt20k) vs (1#20-500k)   |       1.61       0.03    46.59    0.00         1.55        1.68
(3#20-500k) vs (1#20-500k)   |       1.65       0.03    50.32    0.00         1.59        1.72
 (3#gt500k) vs (1#20-500k)   |       1.71       0.04    45.24    0.00         1.64        1.79
   (2#lt20k) vs (1#gt500k)   |       0.73       0.04    19.50    0.00         0.66        0.81
 (2#20-500k) vs (1#gt500k)   |       0.83       0.04    22.78    0.00         0.76        0.91
```

```
  (2#gt500k) vs (1#gt500k)  |       0.82       0.04    22.23    0.00         0.75        0.89
   (3#lt20k) vs (1#gt500k)  |       1.58       0.04    39.69    0.00         1.50        1.66
 (3#20-500k) vs (1#gt500k)  |       1.62       0.04    41.79    0.00         1.54        1.69
  (3#gt500k) vs (1#gt500k)  |       1.68       0.04    41.27    0.00         1.60        1.76
  (2#20-500k) vs (2#lt20k)  |       0.10       0.03     3.77    0.00         0.05        0.15
   (2#gt500k) vs (2#lt20k)  |       0.09       0.03     2.85    0.00         0.03        0.15
    (3#lt20k) vs (2#lt20k)  |       0.84       0.03    30.44    0.00         0.79        0.90
  (3#20-500k) vs (2#lt20k)  |       0.88       0.03    30.71    0.00         0.83        0.94
   (3#gt500k) vs (2#lt20k)  |       0.94       0.03    27.85    0.00         0.88        1.01
 (2#gt500k) vs (2#20-500k)  |      -0.01       0.03    -0.47    0.64        -0.07        0.04
  (3#lt20k) vs (2#20-500k)  |       0.74       0.03    25.59    0.00         0.69        0.80
(3#20-500k) vs (2#20-500k)  |       0.78       0.03    29.99    0.00         0.73        0.84
 (3#gt500k) vs (2#20-500k)  |       0.84       0.03    25.80    0.00         0.78        0.91
   (3#lt20k) vs (2#gt500k)  |       0.76       0.03    23.33    0.00         0.69        0.82
 (3#20-500k) vs (2#gt500k)  |       0.80       0.03    25.70    0.00         0.74        0.86
  (3#gt500k) vs (2#gt500k)  |       0.86       0.03    26.32    0.00         0.79        0.92
  (3#20-500k) vs (3#lt20k)  |       0.04       0.03     1.29    0.20        -0.02        0.10
   (3#gt500k) vs (3#lt20k)  |       0.10       0.04     2.77    0.01         0.03        0.17
 (3#gt500k) vs (3#20-500k)  |       0.06       0.03     1.71    0.09        -0.01        0.13
----------------------------------------------------------------------------------------------

//this is for polSay
FIRST JUST ONLY RELEVANT ONES, and below that all
----------------------------------------------------------------------------------------------
                           |            Delta-method    Unadjusted           Unadjusted
                           |   Contrast   std. err.      z    P>|z|     [95% conf. interval]
---------------------------+------------------------------------------------------------------
                    _at#t3 |
  (1#20-500k) vs (1#lt20k) |      -0.04       0.02    -1.99    0.05        -0.09       -0.00
   (1#gt500k) vs (1#lt20k) |      -0.03       0.03    -1.17    0.24        -0.08        0.02
 (1#gt500k) vs (1#20-500k) |       0.01       0.03     0.54    0.59        -0.04        0.07
  (2#20-500k) vs (2#lt20k) |      -0.01       0.02    -0.45    0.65        -0.04        0.03
   (2#gt500k) vs (2#lt20k) |      -0.01       0.02    -0.33    0.74        -0.04        0.03
 (2#gt500k) vs (2#20-500k) |       0.00       0.02     0.07    0.94        -0.04        0.04
  (3#20-500k) vs (3#lt20k) |      -0.00       0.02    -0.09    0.93        -0.04        0.04
   (3#gt500k) vs (3#lt20k) |      -0.00       0.02    -0.20    0.84        -0.05        0.04
 (3#gt500k) vs (3#20-500k) |      -0.00       0.02    -0.12    0.90        -0.05        0.04
----------------------------------------------------------------------------------------------

. //this is for polSay
. margins t3, at(sf3=(1 2 3)) pwcompare(effects)
Pairwise comparisons of predictive margins              Number of obs = 66,483
Expression: Linear prediction, fixed portion, predict()
1._at: sf3 = 1
2._at: sf3 = 2
3._at: sf3 = 3
----------------------------------------------------------------------------------------------
                           |            Delta-method    Unadjusted           Unadjusted
                           |   Contrast   std. err.      z    P>|z|     [95% conf. interval]
---------------------------+------------------------------------------------------------------
                    _at#t3 |
  (1#20-500k) vs (1#lt20k) |      -0.04       0.02    -1.99    0.05        -0.09       -0.00
   (1#gt500k) vs (1#lt20k) |      -0.03       0.03    -1.17    0.24        -0.08        0.02
    (2#lt20k) vs (1#lt20k) |       0.10       0.02     5.22    0.00         0.06        0.14
  (2#20-500k) vs (1#lt20k) |       0.09       0.02     4.60    0.00         0.05        0.13
   (2#gt500k) vs (1#lt20k) |       0.09       0.02     4.22    0.00         0.05        0.14
    (3#lt20k) vs (1#lt20k) |       0.19       0.02     9.56    0.00         0.15        0.23
  (3#20-500k) vs (1#lt20k) |       0.19       0.02     9.09    0.00         0.15        0.23
   (3#gt500k) vs (1#lt20k) |       0.19       0.02     7.93    0.00         0.14        0.24
 (1#gt500k) vs (1#20-500k) |       0.01       0.03     0.54    0.59        -0.04        0.07
  (2#lt20k) vs (1#20-500k) |       0.14       0.02     6.96    0.00         0.10        0.18
(2#20-500k) vs (1#20-500k) |       0.14       0.02     6.70    0.00         0.10        0.18
 (2#gt500k) vs (1#20-500k) |       0.14       0.02     6.14    0.00         0.09        0.18
  (3#lt20k) vs (1#20-500k) |       0.24       0.02    10.86    0.00         0.19        0.28
(3#20-500k) vs (1#20-500k) |       0.24       0.02    10.97    0.00         0.19        0.28
 (3#gt500k) vs (1#20-500k) |       0.23       0.02     9.62    0.00         0.19        0.28
   (2#lt20k) vs (1#gt500k) |       0.13       0.02     5.37    0.00         0.08        0.18
 (2#20-500k) vs (1#gt500k) |       0.12       0.02     5.10    0.00         0.07        0.17
  (2#gt500k) vs (1#gt500k) |       0.12       0.02     5.22    0.00         0.08        0.17
   (3#lt20k) vs (1#gt500k) |       0.22       0.03     8.91    0.00         0.17        0.27
 (3#20-500k) vs (1#gt500k) |       0.22       0.02     8.93    0.00         0.17        0.27
  (3#gt500k) vs (1#gt500k) |       0.22       0.03     8.56    0.00         0.17        0.27
  (2#20-500k) vs (2#lt20k) |      -0.01       0.02    -0.45    0.65        -0.04        0.03
   (2#gt500k) vs (2#lt20k) |      -0.01       0.02    -0.33    0.74        -0.04        0.03
    (3#lt20k) vs (2#lt20k) |       0.09       0.02     5.55    0.00         0.06        0.13
  (3#20-500k) vs (2#lt20k) |       0.09       0.02     5.10    0.00         0.06        0.13
   (3#gt500k) vs (2#lt20k) |       0.09       0.02     4.24    0.00         0.05        0.13
 (2#gt500k) vs (2#20-500k) |       0.00       0.02     0.07    0.94        -0.04        0.04
  (3#lt20k) vs (2#20-500k) |       0.10       0.02     5.59    0.00         0.07        0.14
(3#20-500k) vs (2#20-500k) |       0.10       0.02     5.82    0.00         0.07        0.13
 (3#gt500k) vs (2#20-500k) |       0.10       0.02     4.67    0.00         0.06        0.14
   (3#lt20k) vs (2#gt500k) |       0.10       0.02     4.94    0.00         0.06        0.14
 (3#20-500k) vs (2#gt500k) |       0.10       0.02     4.97    0.00         0.06        0.14
  (3#gt500k) vs (2#gt500k) |       0.10       0.02     4.76    0.00         0.06        0.14
  (3#20-500k) vs (3#lt20k) |      -0.00       0.02    -0.09    0.93        -0.04        0.04
   (3#gt500k) vs (3#lt20k) |      -0.00       0.02    -0.20    0.84        -0.05        0.04
 (3#gt500k) vs (3#20-500k) |      -0.00       0.02    -0.12    0.90        -0.05        0.04
----------------------------------------------------------------------------------------------
.
```

.19

3 Dropping Countries with Few obs in urban-rural extremes, and Results by Country

Stata apparently estimates coefficients even if there are no observations in a cell, and there are quite a few countries like that where there are very few or no observations in <20k or >500k, for instance Hong Kong is 100% >500k, and there are also countries 0% >500k. Here for robustness/sensitivity check we drop countries if there are fewer than 50obs on <20k or >500k and re-estimate body results and then also provide results by country.

```
use /tmp/allCC.dta,clear
ta cc t3 //whoa!!!
di r(r)
ISO 3166-1 |
   alpha-3 |
   country |   RECODE of town (place size)
      code |     lt20k    20-500k     gt500k |     Total
-----------+---------------------------------+----------
       AND |       730        274          0 |     1,004
       ARG |       200        180        623 |     1,003
       ARM |       545        232        446 |     1,223
       AUS |       453        567        705 |     1,725
       BGD |       991        209          0 |     1,200
       BOL |       625        741        701 |     2,067
       BRA |       185        864        713 |     1,762
       CAN |     1,098      1,694      1,226 |     4,018
       CHL |       152        487        361 |     1,000
       CHN |        50      2,986          0 |     3,036
       COL |       304        722        494 |     1,520
       CYP |       615        385          0 |     1,000
       CZE |       680        370        150 |     1,200
       DEU |       712        616        200 |     1,528
       ECU |       384        466        350 |     1,200
       EGY |       424        456        320 |     1,200
       ETH |       841        380          9 |     1,230
       GBR |       495      1,713        397 |     2,605
       GRC |       420        340        440 |     1,200
       GTM |        71        505        653 |     1,229
       HKG |         0          0      2,075 |     2,075
       IDN |     2,761        346         93 |     3,200
       IRN |       400        609        490 |     1,499
       IRQ |       320        232        648 |     1,200
       JOR |       375        306        522 |     1,203
       JPN |        51        786        516 |     1,353
       KAZ |       457        456        363 |     1,276
       KEN |       499        421        122 |     1,042
       KGZ |       710        290        200 |     1,200
       KOR |         0        364        881 |     1,245
       LBN |       410        660        130 |     1,200
       LBY |       480        544        172 |     1,196
       MAC |         0          0      1,023 |     1,023
       MAR |       380        480        340 |     1,200
       MDV |       738        301          0 |     1,039
       MEX |       858        463        403 |     1,724
       MMR |     1,116         84          0 |     1,200
       MNG |       378        257      1,003 |     1,638
       MYS |       744        428        141 |     1,313
       NGA |       848        375         14 |     1,237
       NIC |       381        709        110 |     1,200
       NIR |        39        407          0 |       446
       NLD |       104      1,710        217 |     2,031
       NZL |       420        387        187 |       994
       PAK |     1,272        249        474 |     1,995
       PER |       355        945        100 |     1,400
       PHL |       910        290          0 |     1,200
       PRI |        73      1,054          0 |     1,127
       ROU |       703        435        119 |     1,257
       RUS |       594        626        590 |     1,810
       SGP |         0          0      2,012 |     2,012
       SRB |       403        447        196 |     1,046
       SVK |       750        450          0 |     1,200
       THA |     1,244        116        140 |     1,500
       TJK |       880        210        110 |     1,200
       TUN |       514        614         80 |     1,208
       TUR |       240      1,767        408 |     2,415
       TWN |        40      1,165         18 |     1,223
       UKR |       638        355        296 |     1,289
       URY |       270        290        440 |     1,000
       USA |         0      1,443        385 |     1,828
       VEN |       180        661        349 |     1,190
       VNM |         0      1,049        151 |     1,200
       ZWE |       945        270          0 |     1,215
-----------+---------------------------------+----------
     Total |    32,455     37,238     23,306 |    92,999
levelsof cc,loc(__cc)
foreach _cc in `__cc'{
count if t3 == 3 & cc == "`_cc'"
drop if `r(N)'<50 & cc == "`_cc'"
count if t3 == 1 & cc == "`_cc'"
drop if `r(N)'<50 & cc == "`_cc'" //extra robustness check
di "`_cc'"  `r(N)'
}
ta cc t3
di r(r)
. ta cc t3
ISO 3166-1 |
   alpha-3 |
   country |   RECODE of town (place size)
```

code	lt20k	20-500k	gt500k	Total
ARG	200	180	623	1,003
ARM	545	232	446	1,223
AUS	453	567	705	1,725
BOL	625	741	701	2,067
BRA	185	864	713	1,762
CAN	1,098	1,694	1,226	4,018
CHL	152	487	361	1,000
COL	304	722	494	1,520
CZE	680	370	150	1,200
DEU	712	616	200	1,528
ECU	384	466	350	1,200
EGY	424	456	320	1,200
GBR	495	1,713	397	2,605
GRC	420	340	440	1,200
GTM	71	505	653	1,229
IDN	2,761	346	93	3,200
IRN	400	609	490	1,499
IRQ	320	232	648	1,200
JOR	375	306	522	1,203
JPN	51	786	516	1,353
KAZ	457	456	363	1,276
KEN	499	421	122	1,042
KGZ	710	290	200	1,200
LBN	410	660	130	1,200
LBY	480	544	172	1,196
MAR	380	480	340	1,200
MEX	858	463	403	1,724
MNG	378	257	1,003	1,638
MYS	744	428	141	1,313
NIC	381	709	110	1,200
NLD	104	1,710	217	2,031
NZL	420	387	187	994
PAK	1,272	249	474	1,995
PER	355	945	100	1,400
ROU	703	435	119	1,257
RUS	594	626	590	1,810
SRB	403	447	196	1,046
THA	1,244	116	140	1,500
TJK	880	210	110	1,200
TUN	514	614	80	1,208
TUR	240	1,767	408	2,415
UKR	638	355	296	1,289
URY	270	290	440	1,000
VEN	180	661	349	1,190
Total	23,769	25,752	16,738	66,259

```
. di r(r)
44
```

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3.1 Robustness/Sensitivity Check Dropping Countries With Fewer Than 50obs in <20k or >500k

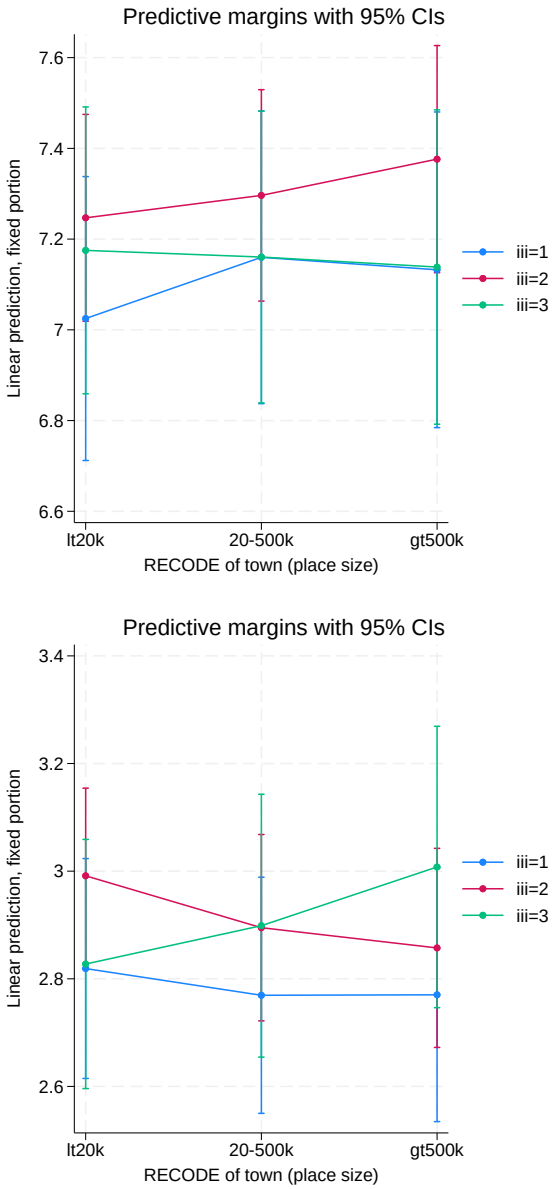


Figure 6: As stated in the body, nothing changes substantively here, the results from the body are robust when dropping countries where >500k <50obs and <20k <50obs.

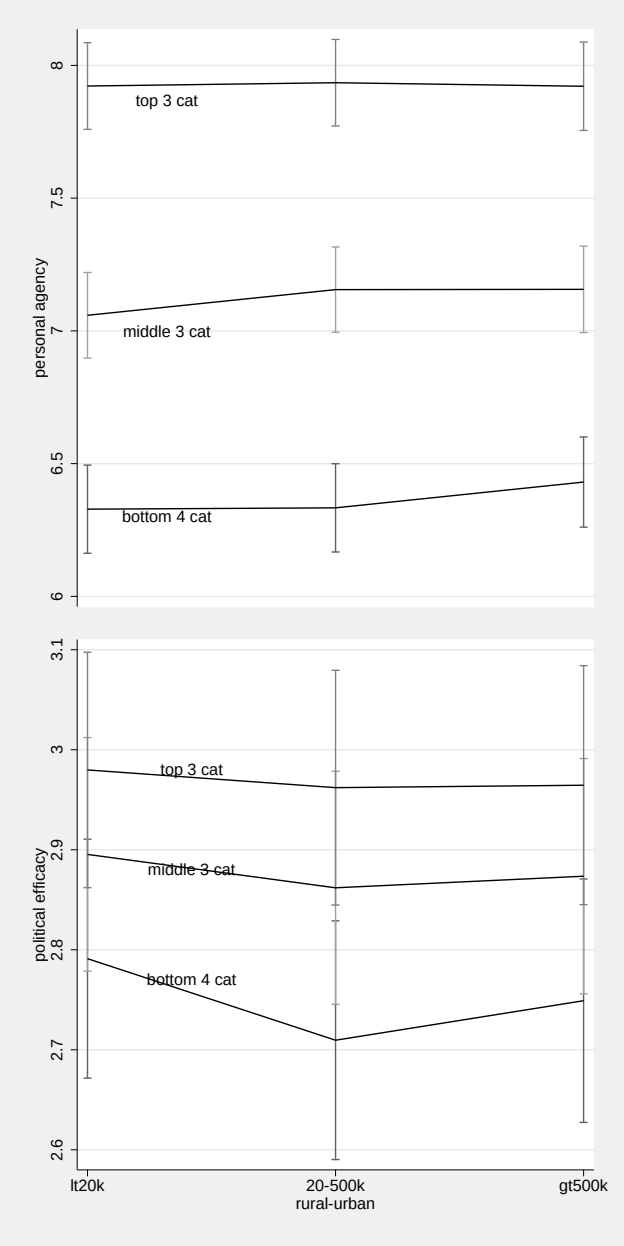


Figure 7: zzR As stated in the body, these results are robust when dropping countries where >500k <50obs only (next figure); but not here, when dropping when dropping countries where >500k <50obs and <20k <50obs.

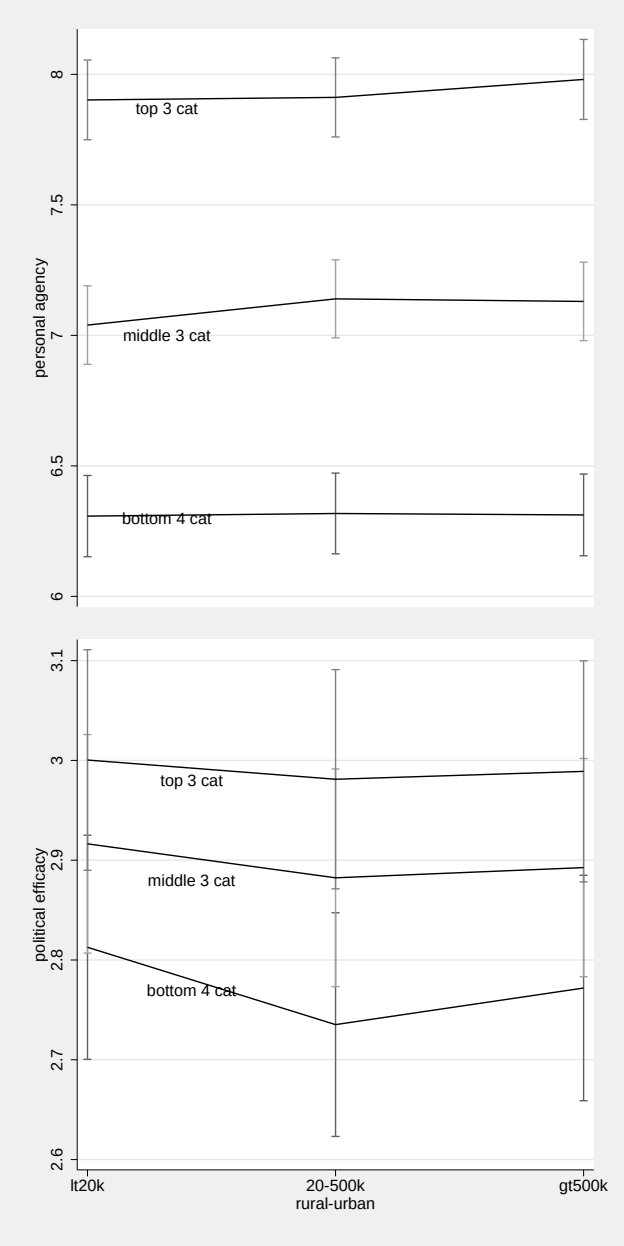


Figure 8: zzR2 As stated in the body, these results are robust when dropping countries where >500k <50obs only (this figure); but not in previous figure, when dropping when dropping countries where >500k <50obs and <20k <50obs.

3.2 Results by Country

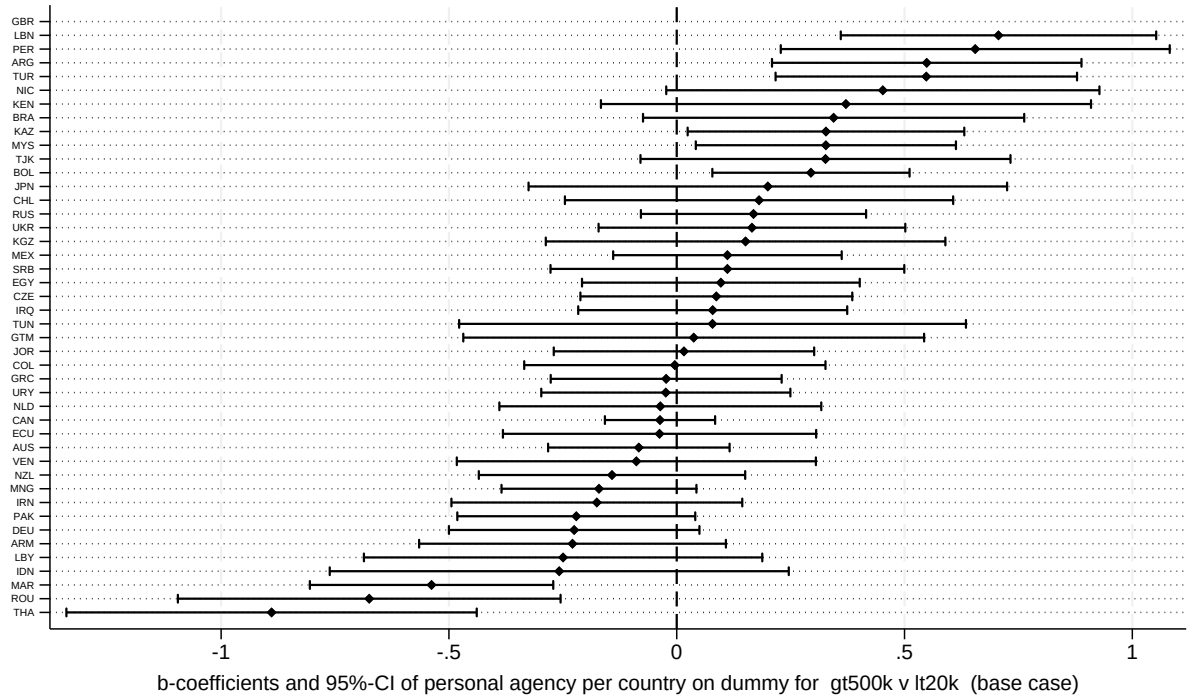


Figure 9: ts-freeR

In figure 9 it is difficult to see any pattern, except perhaps Latin countries such as PER, ARG, and marginally BRA tend to have more personal agency in cities; most countries are insignificant.

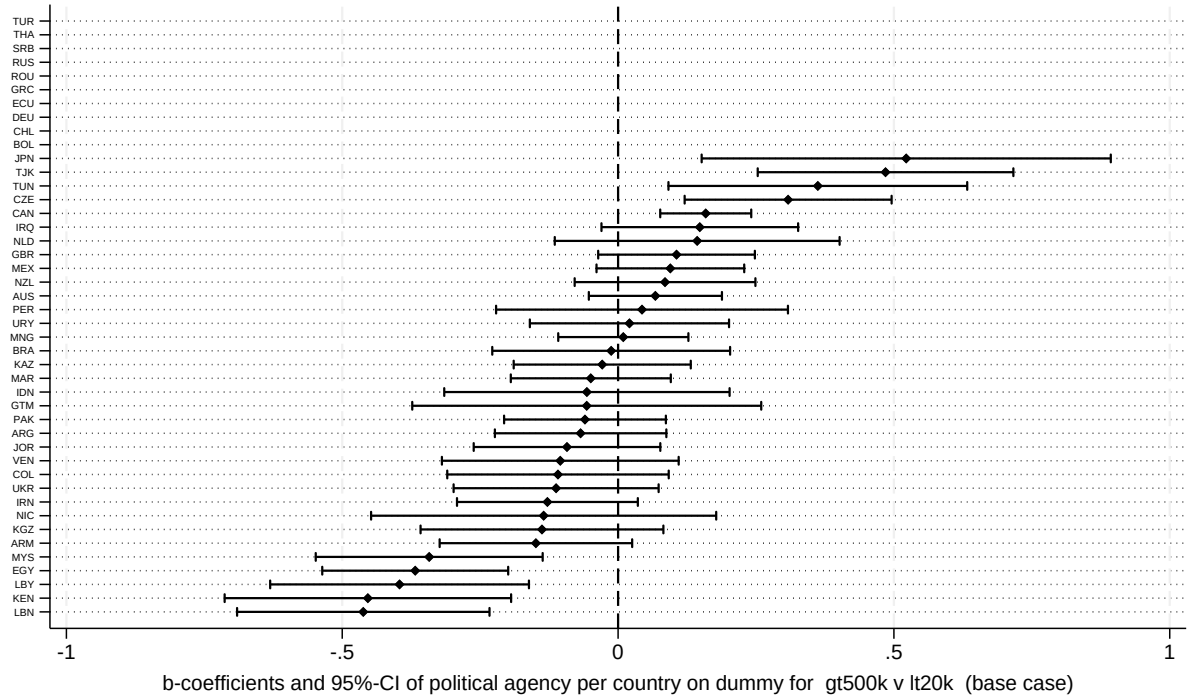


Figure 10: ts-polSayR

in Figure 10 there is some pattern as expected–rich countries such as JPN, CZE, and CAN have higher political agency in cities, and poor countries such as LBN, KEN, and LBY have lower. Still most countries are insignificant.